

CHAPTER 4

DYNAMICS: NEWTON'S LAWS OF MOTION



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Physics I_PNAP0123

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AIM: to determine the cause of a given motion; to relate a given motion with its cause

LEARNING OUTCOMES:

At the end of the lesson, students should be able to

- State each of three Newton's Laws of Motion
- Explain by illustrating real-life situations each of Newton's Laws of Motion
- Explain the concepts of mass and weight; explain normal force, friction force
- Apply the second law of motion and free-body diagram to solve problems involving forces, including of motion on an incline

Sections of Chapter 4

- **Force**
- **Newton's First Law of Motion**
- **Mass**
- **Newton's Second Law of Motion**
- **Newton's Third Law of Motion**
- **Weight – the Force of Gravity; Normal Force**

Sections of Chapter 4

- **Solving Problems with Newton's Laws:
Free-Body Diagrams**
- **Applications Involving Friction, Inclines**
- **Problem Solving – A General Approach**

4-1 Force

Mechanics (studies on motion) consists of two parts:

- 1. Kinematics**
- 2. Dynamics**

What is dynamics?

Dynamics relates a given motion with the cause of that motion.

Dynamics of any motion is associated with **force.**

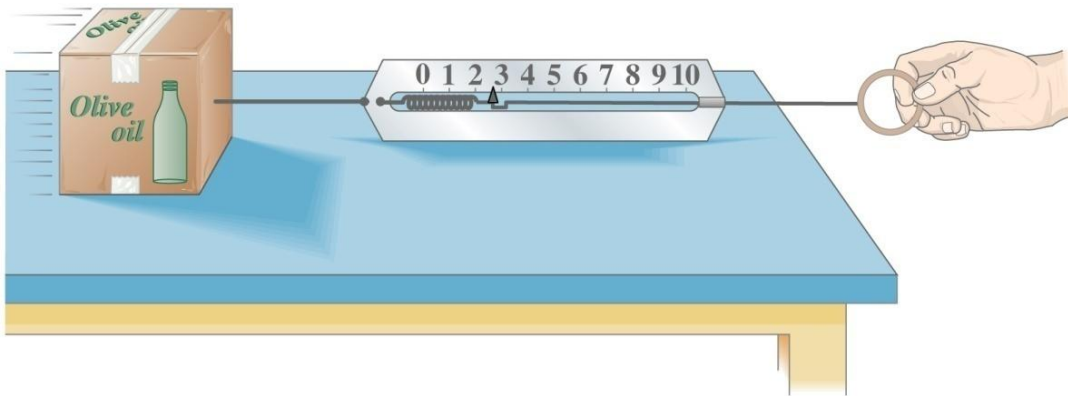
But... *what is force?*

4-1 Force

A force is a push or a pull.

Force is needed in order to:

- ▶ move an object from rest
- ▶ change the speed of a moving object
- ▶ change the direction of a moving object

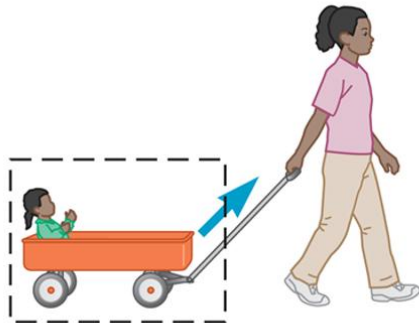
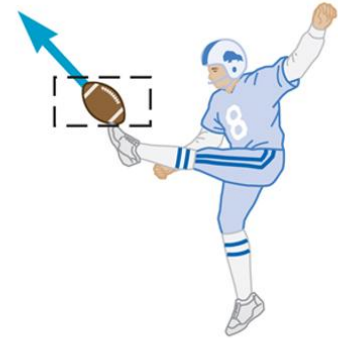
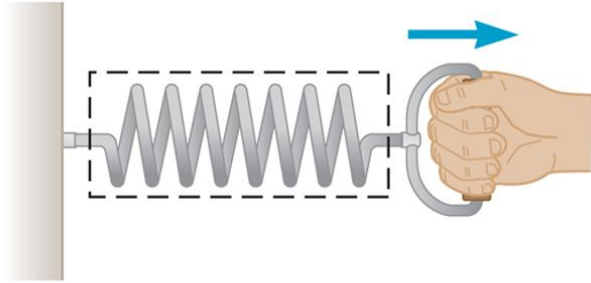


Magnitude of a force can be measured by using a spring scale.

4-1 Force

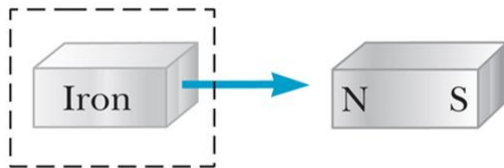
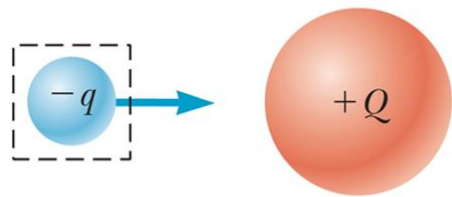
- Force is a measure of pulling or pushing interaction between two objects.
- It is a vector quantity: has ***magnitude and direction***
- There are 2 classes of forces:
 - i. contact force
 - ii. field force

Contact forces result from physical contact between two objects

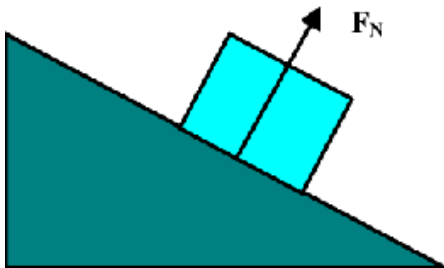
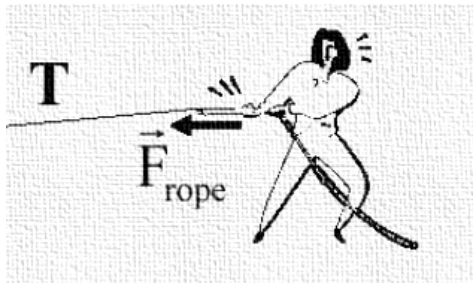
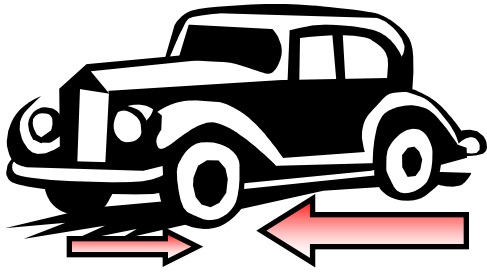


Field forces act between objects separated over a distance.

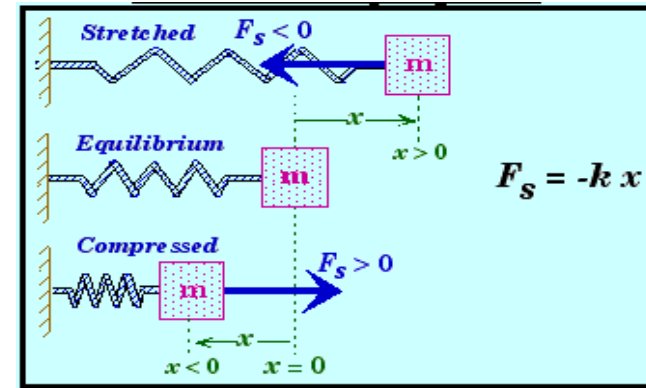
Also called “*action at a distance*”



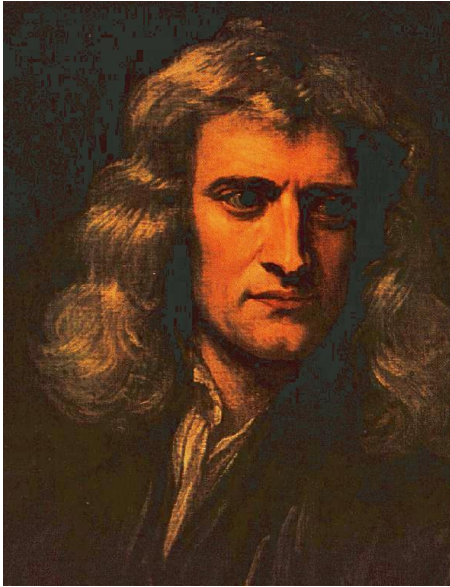
4-1 Force



- Gravitational Force
 - Buoyant Force
 - Friction Force
 - Tension Force
 - Spring Force
 - Normal Force
- etc.*



Background



Sir Isaac Newton (1643-1727) an English scientist and mathematician famous for his discovery of the law of gravity also discovered the **three laws of motion**.

He published them in his book *Philosophiae Naturalis Principia Mathematica* (mathematic principles of natural philosophy) in 1687. Today these laws are known as **Newton's Laws of Motion** and describe the motion of all objects on the scale we experience in our everyday lives.

4-2 Newton's First Law of Motion

An object at rest tends to stay at rest;
an object in motion tends to remain in motion
with constant speed in the same direction;
as long as the net force is zero

The 1st law

The law of inertia

Objects keep on
doing what
they're doing.



4-2 Newton's First Law of Motion

- The net force is defined as *the vector sum of all the forces* acting on the object.
- When all the forces are balanced i.e. the net force is zero, the object can be in 2 states, either **stationary**, or **move with constant velocity (at a constant speed in a straight line direction)**.
- Under this condition, the acceleration (a) of the object must be zero, thus:
 - Object at rest: $v = 0$ and $a = 0$
 - Object in motion: $v \neq 0$ and $a = 0$

4-2 Newton's First Law of Motion

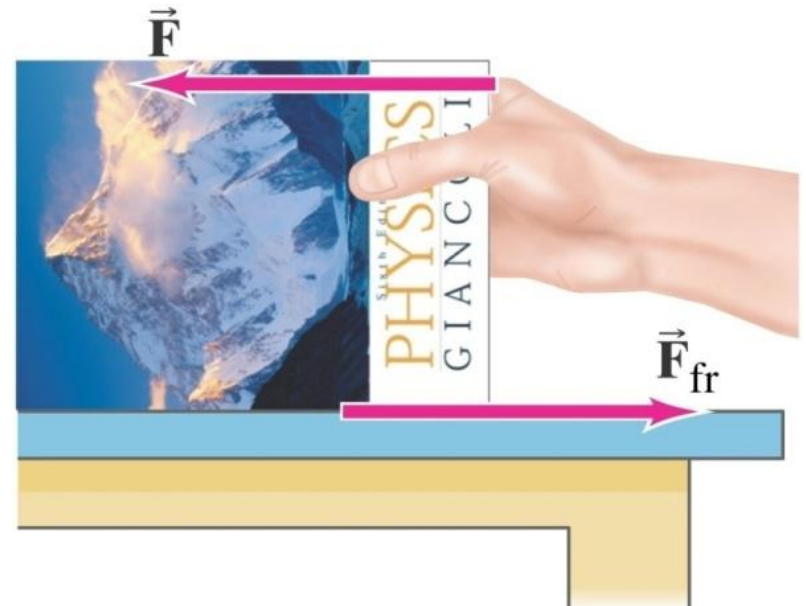
No unbalanced force in Newton's 1st Law

1) The boy is standing still

The forces on the person are balanced.



2) Physics book moving at a constant speed across a table



QUICK QUIZ 4-1

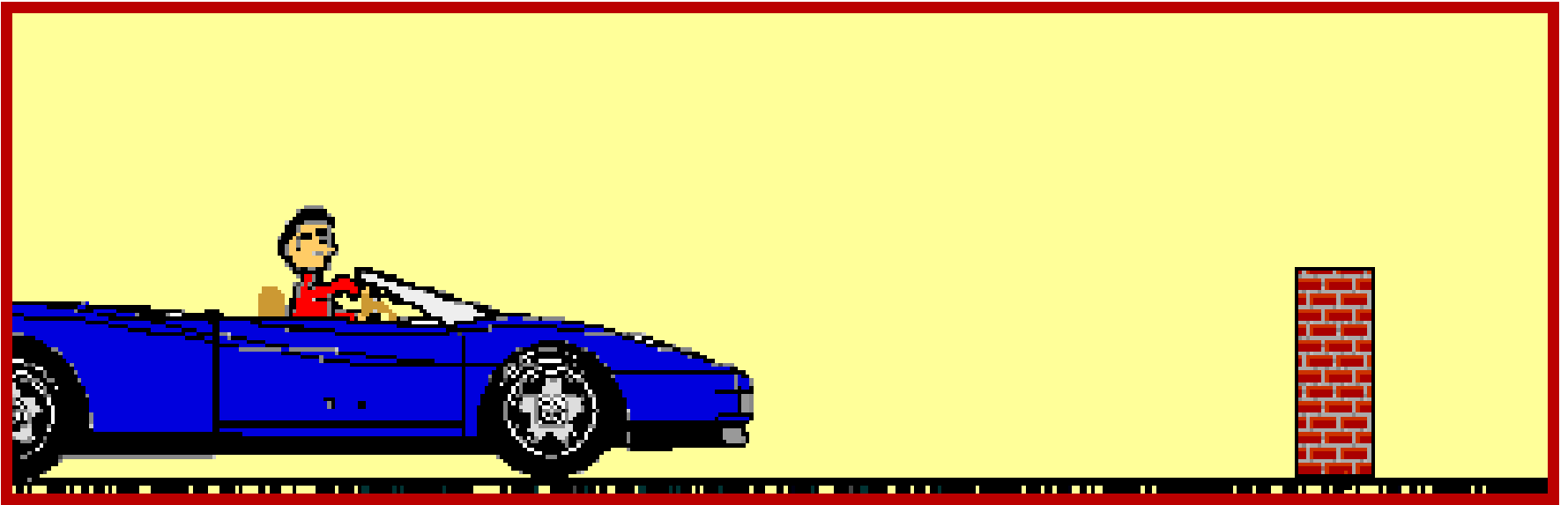
An object moves with a constant velocity. Which of the following cannot be true for the object?

- A. No forces act on the object.
- B. A single force acts on the object.
- C. Two forces act on the object.
- D. Three forces act on the object.

4-2 Newton's First Law of Motion

The 1st law and inertia

The car and the wall

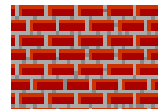


This law is the very reason why you should always wear your seatbelt.

4-2 Newton's First Law of Motion

The 1st law and inertia

The motorcyclist



4-2 Newton's First Law of Motion

The 1st law and inertia

The Truck and the Ladder



4-2 Newton's First Law of Motion

Inertial Reference Frames

An *inertial reference frame* is one in which Newton's first law is valid.

Accelerating and rotating frames are NOT inertial reference frames.

Test your Understanding 1

When the school bus driver steps on the brake to bring the bus to a stop, all the backpacks lying on the floor slide forward.
What is the force that cause it?

4-2 Newton's First Law of Motion

Distinction between Mass and Weight

- **Mass, m** is generally defined as quantity of matter, which is different from weight, w .
- Mass is an intrinsic property of an object, while weight is not.
- **Weight** is the force exerted on the object by the earth's gravity:

$$F = mg$$

where, m = mass

g = gravity acceleration = 9.80 m/s^2

- On the moon, the gravity acceleration is $\sim 1/6 g$, so the same mass will weigh much less.

4-2 Newton's First Law of Motion

Mass and Inertia

- **Inertia is the tendency of an object to remain at rest or in motion at a constant velocity.**
- **The more mass an object has, the more inertia it has, since the harder it is to change its motion.**
- **Therefore, **mass** of an object is more accurately defined as a quantitative measure of inertia.**

4-2 Newton's Second Law of Motion

The acceleration, a of an object is directly proportional to the net force, F_{net} acting on it, and is inversely proportional to its mass, m .

The direction of acceleration is in the direction of the force.

The 2nd law

$$\Sigma \vec{F} = m\vec{a}$$

where $\vec{F}$ = force,

$\vec{a}$ = acceleration

More about..

$$F = ma$$

If you **double the mass**, you **double the force**. If you **double the acceleration**, you **double the force**.

What if you **double both the mass** and **the acceleration**?

$$(2m)(2a) = 4F$$

Doubling the mass *and* the acceleration **quadruples the force**.

So . . what if you **decrease the mass by half**?

How much force would the object have now?

4-2 Newton's Second Law of Motion

$$\Sigma \vec{F} = m\vec{a}$$

Newton's Second Law

The acceleration of an object increases with increased force, decreases with increased mass, and is in the same direction as the force.

Increasing Force Increases Acceleration



small force



acceleration



larger force



acceleration

The force exerted on the cart by the man is greater than the force exerted on the same cart by the boy, so the acceleration is greater.

Increasing Mass Decreases Acceleration



small mass



acceleration



larger mass



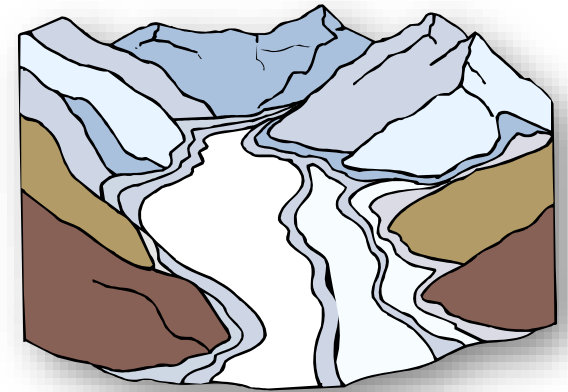
acceleration

The mass of the full cart is greater than the mass of the empty cart, and the boy is pushing with the same force, so the acceleration is less.

What does $F = ma$ says...

It basically means that the force of an object comes from its mass and its acceleration.

Something very massive (high mass) that's changing speed very slowly (low acceleration), like a glacier, can still have a great force.



Something very small (low mass) that's changing speed very quickly (high acceleration), like a bullet, can still have a great force. That's why bullet is deadly, it is *the power of mass moving at high speed!*



4-2 Newton's Second Law of Motion

Force is a vector, thus $\Sigma \vec{F} = m\vec{a}$ can be applied separately along each coordinate axis.

Unit of force in the SI system is **newton (N)**.

TABLE 4-1
Units for Mass and Force

System	Mass	Force
SI	kilogram (kg)	newton (N) (= kg · m/s ²)
cgs	gram (g)	dyne (= g · cm/s ²)
British	slug	pound (lb)
Conversion factors: 1 dyne = 10 ⁻⁵ N; 1 lb ≈ 4.45 N.		

Note that the pound is a unit of force, not mass, and therefore is equated to newtons, not to kilograms.

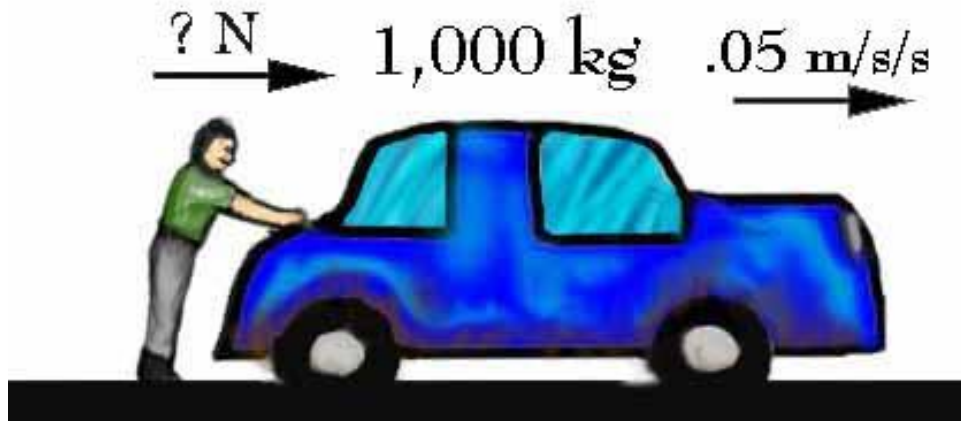
QUICK QUIZ 4-2

An object accelerates when it experiences a non-zero net force. Which of the following statements is always TRUE?

- A. The object moves in the direction of the net force.
- B. The object's velocity increases.
- C. The object's velocity is in the same direction as its acceleration.
- D. The object's acceleration is in the same direction as the net force that acts upon it.

Example 1

Ahmad's car, which weighs 1,000 kg, is out of gas. Ahmad is trying to push the car to a gas station, and he makes the car go 0.05 m/s^2 . Using Newton's Second Law, compute how much force Ahmad is applying to the car.



Newton's 2nd Law

$$F = ma$$

$$F = 1,000 \times 0.05$$

$$= 50 \text{ Newtons}$$

4-2 Newton's Third Law of Motion

Whenever an object exerts a force on another object, the second object exerts a force back on the first object which is equal in magnitude but opposite in direction.

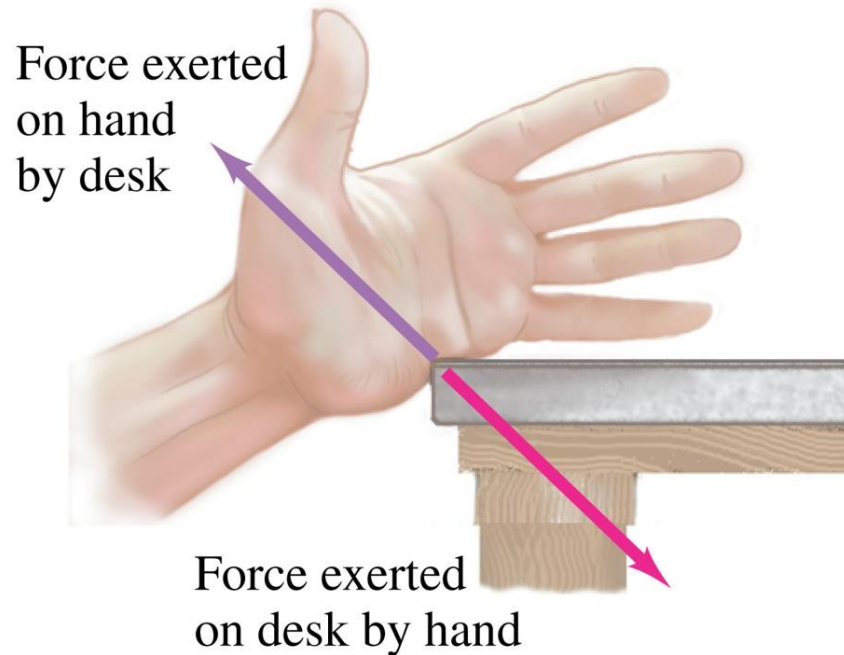
The 3rd law

'For every action, there is an equal but opposite reaction'



- Equivalent to saying a single isolated force cannot exist!

4-2 Newton's Third Law of Motion



**Your hand push the edge of the desk.
The desk push back on your hand causing your
hand to be distorted!**

Forces



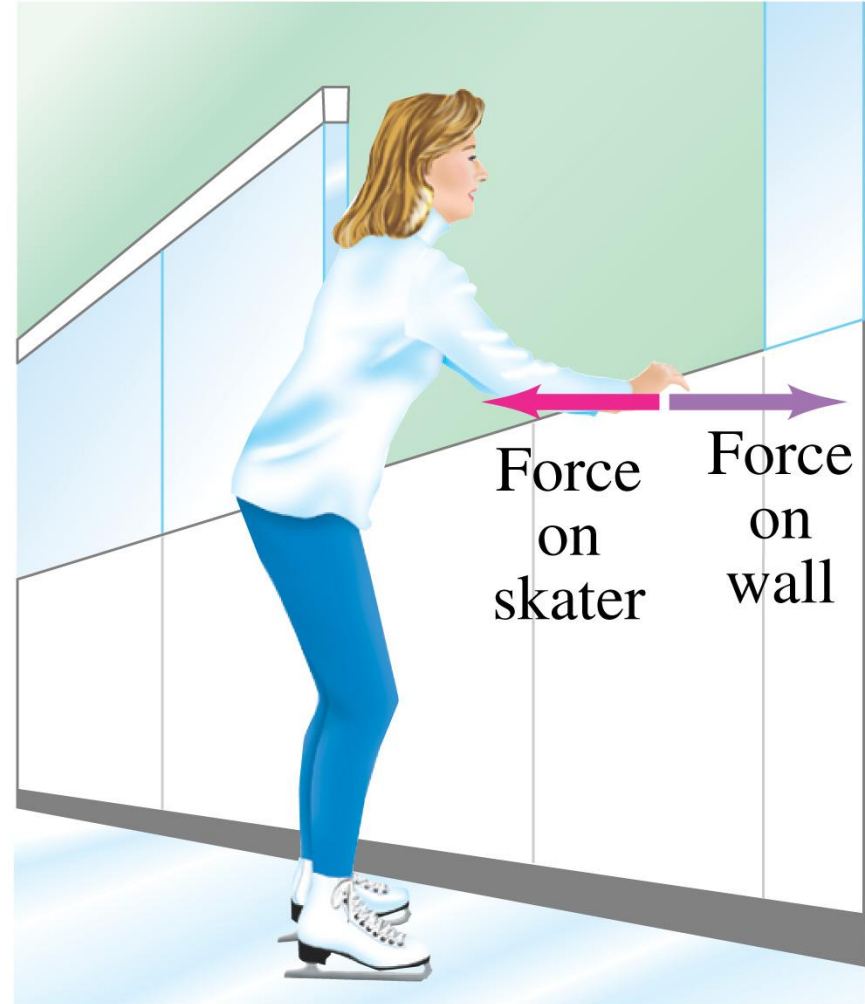
Juergen Hasenkopf/Alamy Stock Photo

4-2 Newton's Third Law of Motion

The wall exerts a force back on the ice skater causing her to move backward.

**** Note that:**

the action-reaction forces are exerted on different objects



4-2 Newton's Third Law of Motion

Rocket propulsion :

hot gases from combustion spew out of the tail of the rocket at high speeds. The reaction force is what propels the rocket.

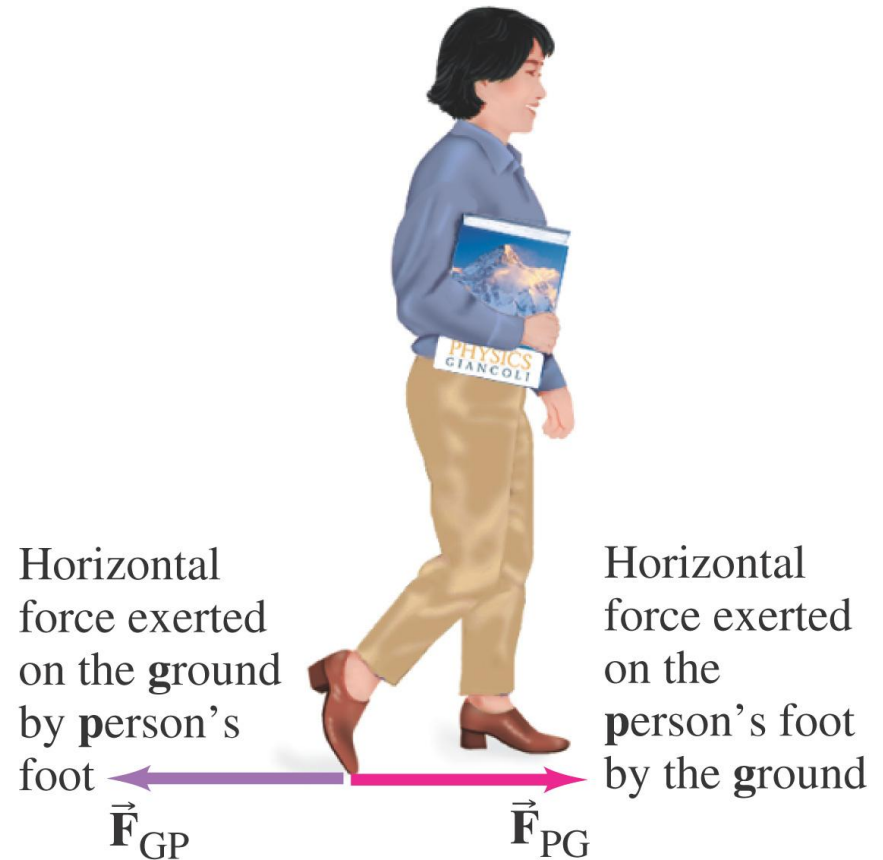


4-2 Newton's Third Law of Motion

Use double subscript notation:

the first subscript is the object that the force is being exerted on; the second is the source.

$$\vec{F}_{GP} = -\vec{F}_{PG}$$



QUICK QUIZ 4-3

When you sit in a chair there is necessarily a reaction force to the gravitational force acting on your body. Which of the following best describes this reaction force?

- A. the force of gravity you exert on the Earth
- B. the force you exert downward on the seat of the chair
- C. the normal force exerted by the chair onto you
- D. none of these forces

4-2 Newton's Third Law of Motion

Test your Understanding 2

In an accident, a truck collides head-on with a kancil. Which experience the greater amount of force?

Test your Understanding 3

A truck collides head-on with a kancil. Which experience the greater acceleration?

QUICK QUIZ 4-4

A race car moving quickly around a track collides with a mosquito, which splatters on the car's windshield. Which object experiences a greater magnitude impact force?

- A. Both experience the same force.
- B. One cannot tell; you need to know the speeds of the mosquito and the race car.
- C. the mosquito
- D. the race car

QUICK QUIZ 4-5

A race car moving quickly around a track collides with a mosquito, which splatters on the car's windshield. Which object experiences a greater magnitude acceleration?

- A. Both experience the same acceleration.
- B. the race car
- C. the mosquito
- D. One cannot tell; you need to know the speeds of the mosquito and the race car.

QUICK QUIZ 4-6

A model rocket of mass m is fired upward from a launch pad with some initial speed, v . A gravitational force is exerted on the rocket

- A. at all points in its motion except at its highest point.
- B. at all points in its motion.
- C. at no points in its motion.
- D. at none of these choices.

4-3 Weight – the Force of Gravity

Weight is the force exerted on an object by gravity.

Close to the surface of the Earth, where the gravitational force is nearly constant, the weight F_G is:

$$\vec{F}_G = m\vec{g}$$

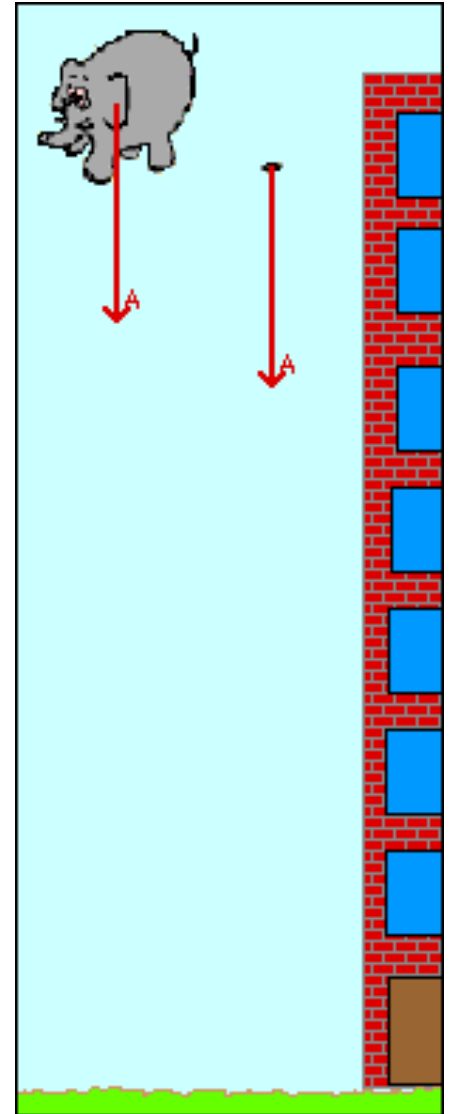
where m = mass of the object,
 g = gravity acceleration = 9.80 m.s^{-2}

4-3 Weight – the Force of Gravity

Concepts of weight, mass & gravity acceleration

***Free Fall:
Elephant and Feather***

Which object will hit the ground first?



TRUE OR FALSE?...

1. The elephant and the feather each have the same force of gravity.
2. The elephant has more mass, yet both elephant and feather experience the same force of gravity.
3. The elephant experiences a greater force of gravity, yet both the elephant and the feather have the same mass.
4. On earth, all objects (whether an elephant or a feather) have the same force of gravity.
5. The elephant weighs more than the feather, yet they each have the same mass.
6. The elephant clearly has more mass than the feather, yet they each weigh the same.
7. The elephant clearly has more mass than the feather, yet the amount of gravity (force) is the same for each.
8. The elephant has the greatest acceleration, yet the amount of gravity is the same for each.

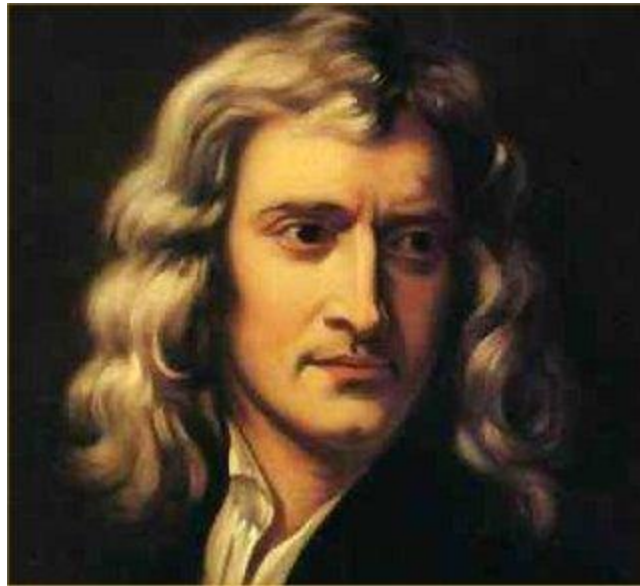
1666, England

Legend has it that Isaac Newton formulated gravitational theory in 1665 or 1666 after watching an apple fall and asking why the apple fell straight down, rather than sideways or even upward.



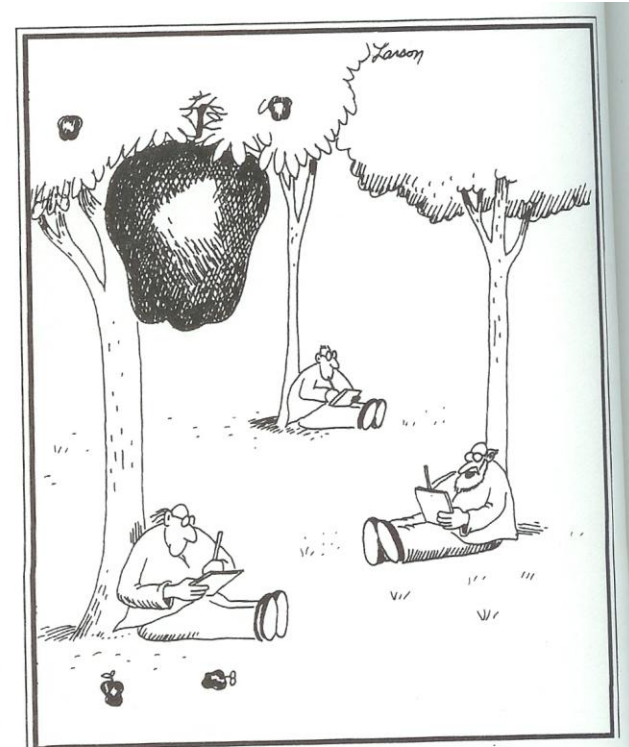
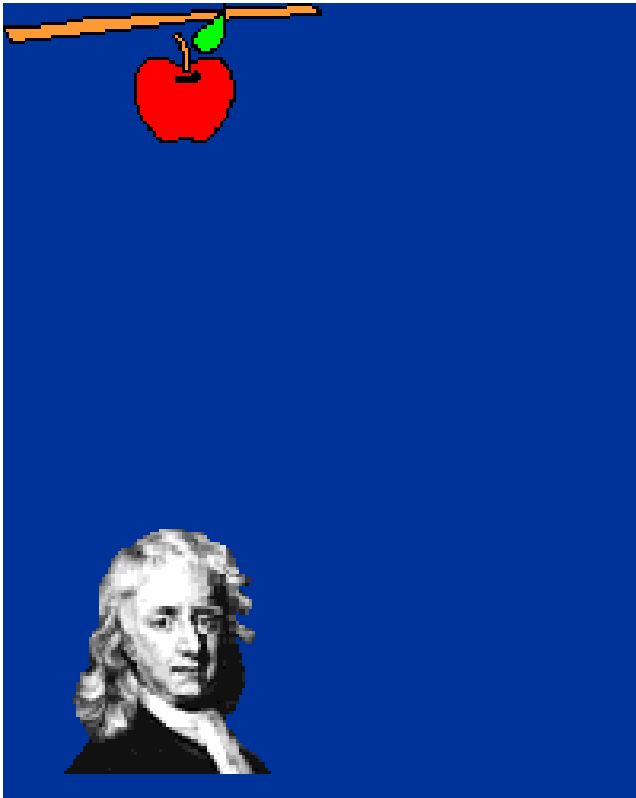
Isaac Newton (1643-1727)

Mathematician, physicist



4-3 Weight - the Force of Gravity

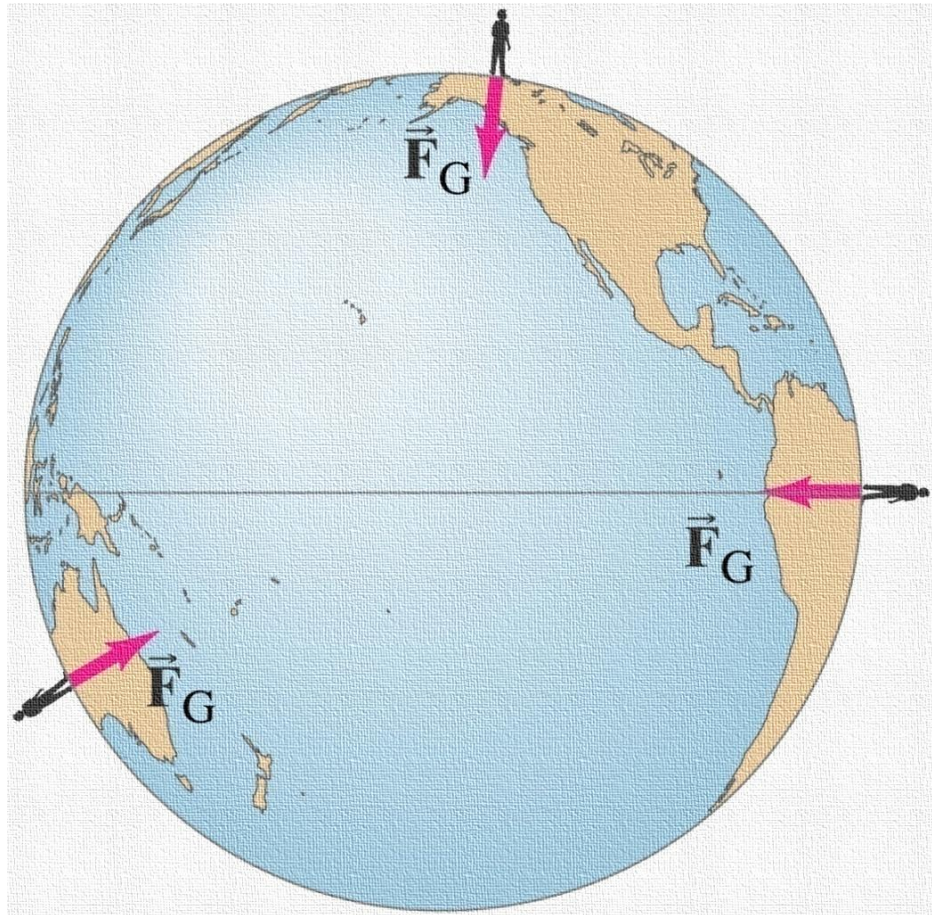
The apple was attracted to the earth



"Nothing yet. ... How about you, Newton?"

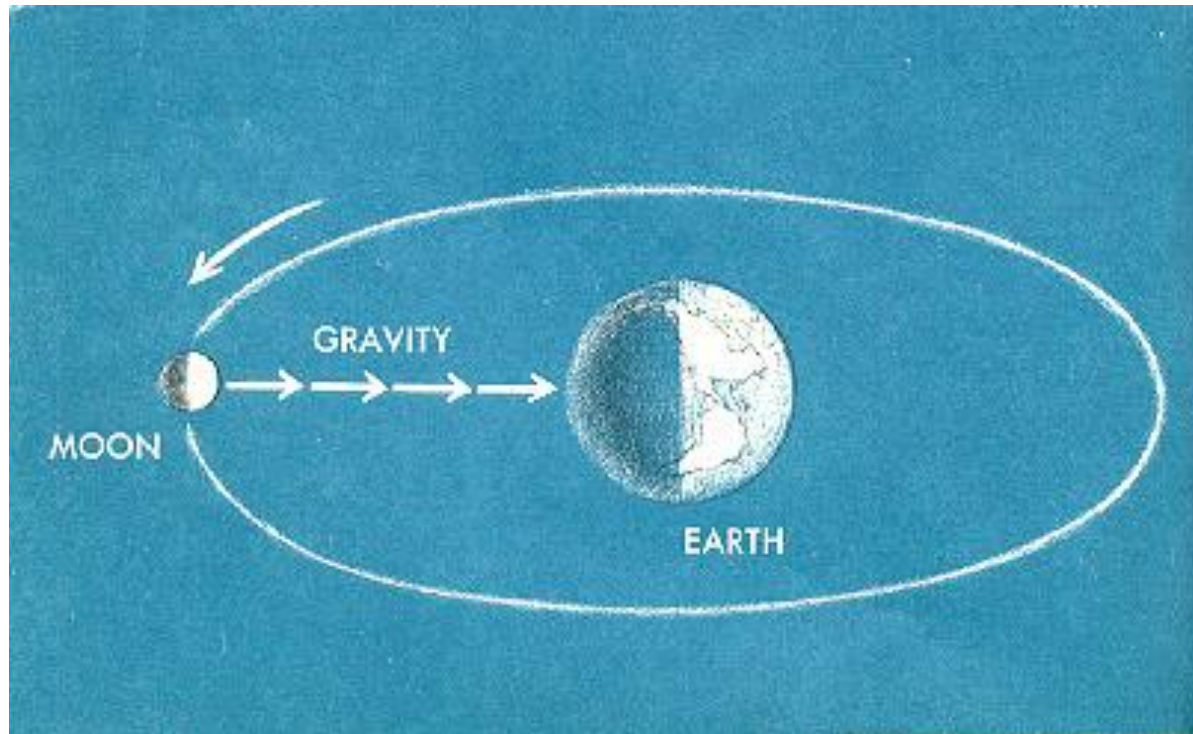
4-3 Weight - the Force of Gravity

What exerts the force of gravity on all objects on the earth?



Some things to ponder..

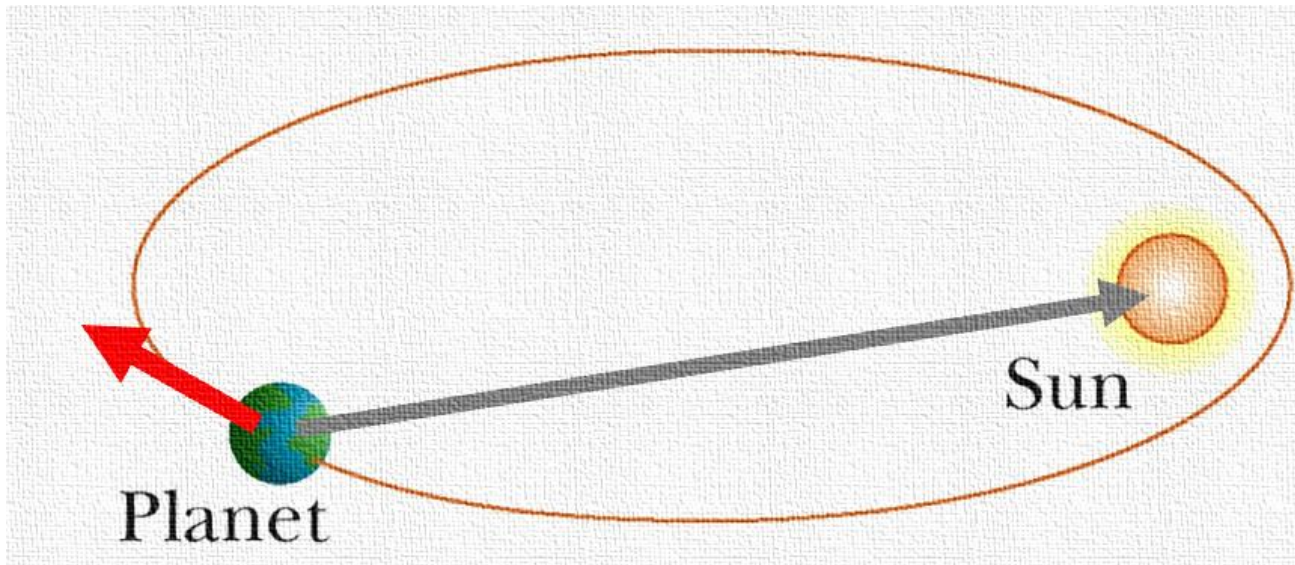
- If the planets are orbiting the sun, **what force** is keeping them in orbit?
- **What force** keeps the moon in its orbit?
- Could the force of gravity be **universal**?



4-3 Weight - the Force of Gravity

Newton realized that the force must be due to the Earth.

This led him to believe it is the same force of gravity which keeps the Moon in its orbit, and the planets to orbit around the sun!



4-3 Weight – the Force of Gravity

Newton conclude that the gravitational force must be proportional to both masses.

Observation of planetary orbits led him conclude that the force decreases as the inverse of the square of the distance between the masses.

This led him to formulate the relationship for the gravitational force:

4-3 Weight – the Force of Gravity

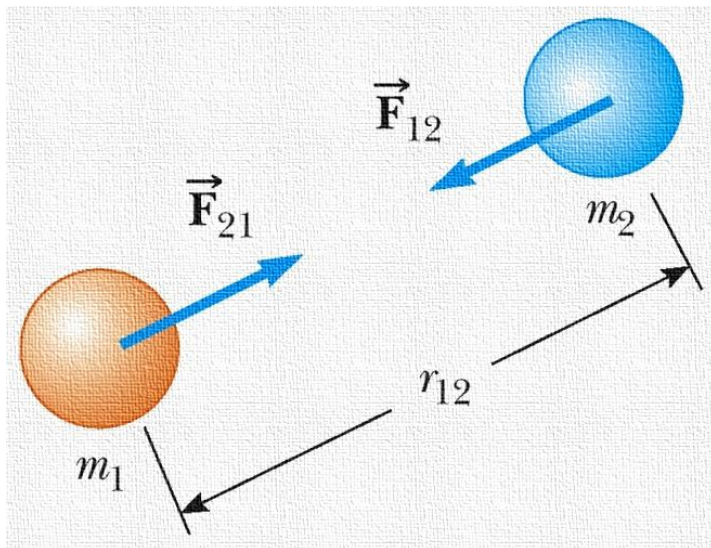
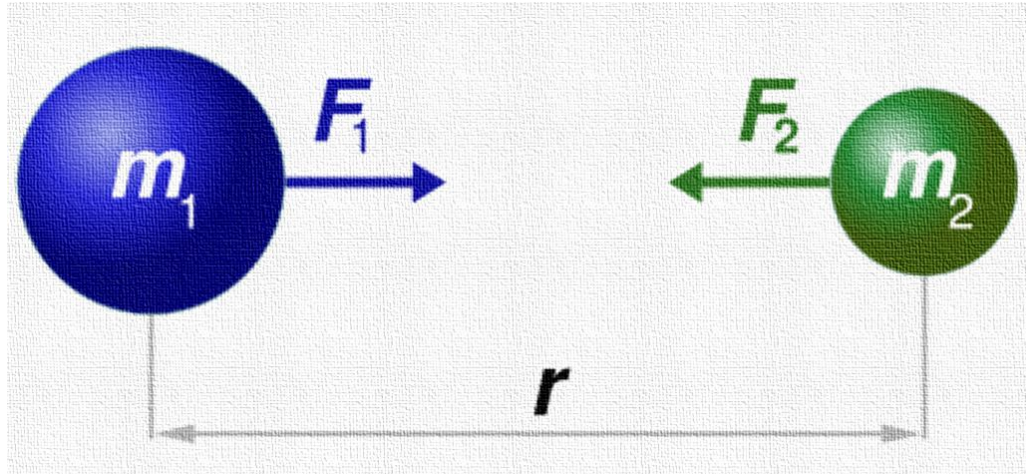
The Law of Universal Gravitation

$$F_G \propto \frac{m_1 m_2}{r^2}$$

Any two objects in the universe attract each other with a gravitational force that is proportional to the product of their masses, and inversely proportional to the square of the distance between them.

The force acts along the line joining the center of mass of the two objects.

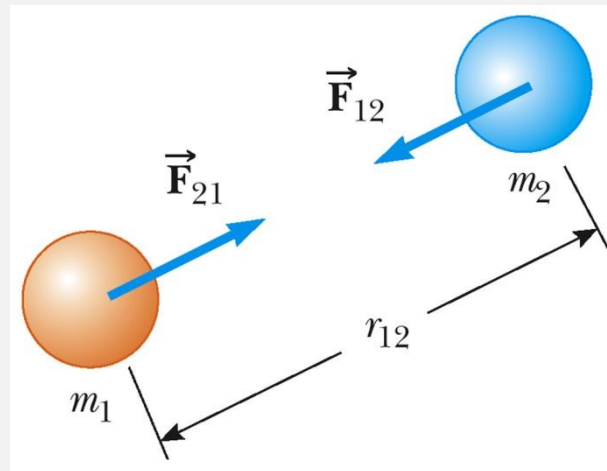
Newton's Law of Universal Gravitation



$$F_1 = F_2 = G \frac{m_1 \times m_2}{r^2}$$

Newton's Law of Universal Gravitation

- The force that mass 1 exerts on mass 2 is equal and opposite to the force mass 2 exerts on mass 1
- The forces form a Newton's third law action-reaction



**** The gravitational force exerted by a uniform sphere on a particle outside the sphere is the same as the force exerted if the entire mass of the sphere were concentrated at its center**

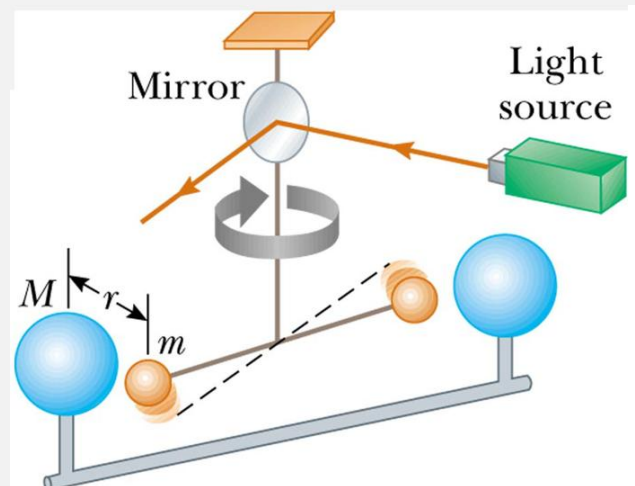
Newton's Law of Universal Gravitation

$$F = G \frac{m_1 m_2}{r^2}$$

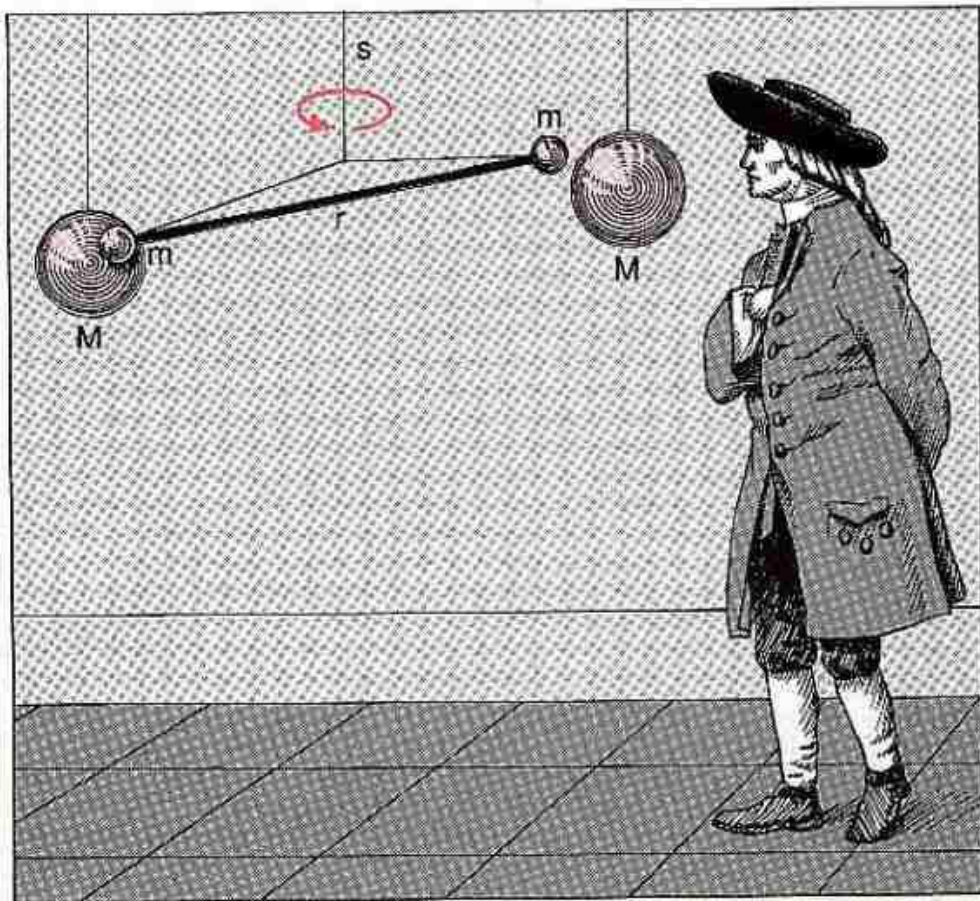
- G is the gravitational constant:

$$G = 6.673 \times 10^{-11} \text{ N m}^2 / \text{kg}^2$$

- This is an example of an *inverse square law*
- Determined experimentally by Henry Cavendish in 1798

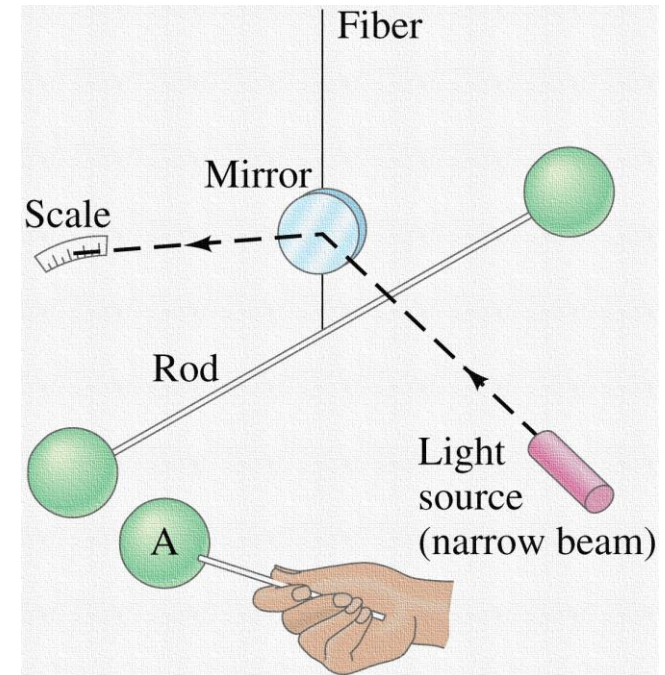


Henry Cavendish's experiment determined the proportionality constant G in 1798.



All diagrams by Peter Gardiner

Henry Cavendish with the famous torsion balance experiment that determined the gravitational constant G and demonstrated Newton's inverse-square law of gravitation. Large lead spheres placed close to small ones caused angular deflections



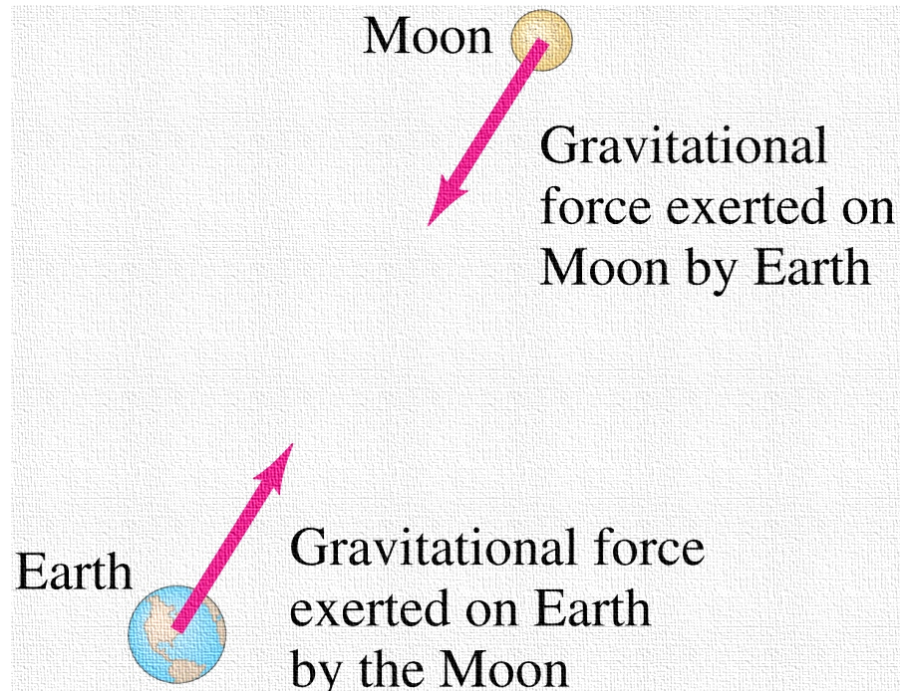
Sensitive torsion
balance apparatus

5-6 Newton's Law of Universal Gravitation

Newton's Third Law:

Earth exerts a pulling force on the moon;
the moon exerts an equal but opposite
direction of force on the Earth.

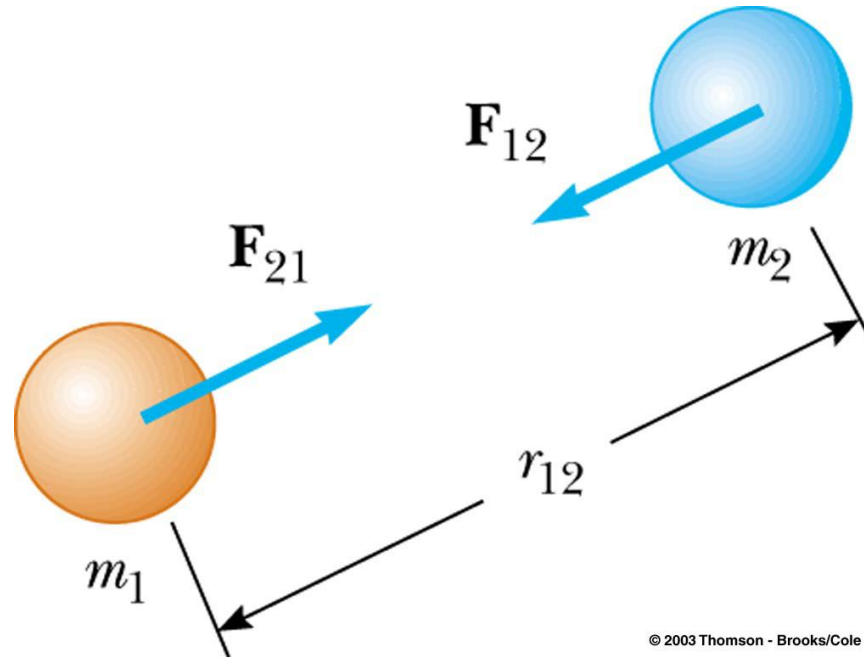
In this case, due to
disparity in masses, the
reaction force i.e. force
on the earth due to the
moon is undetectable.



5-6 Newton's Law of Universal Gravitation

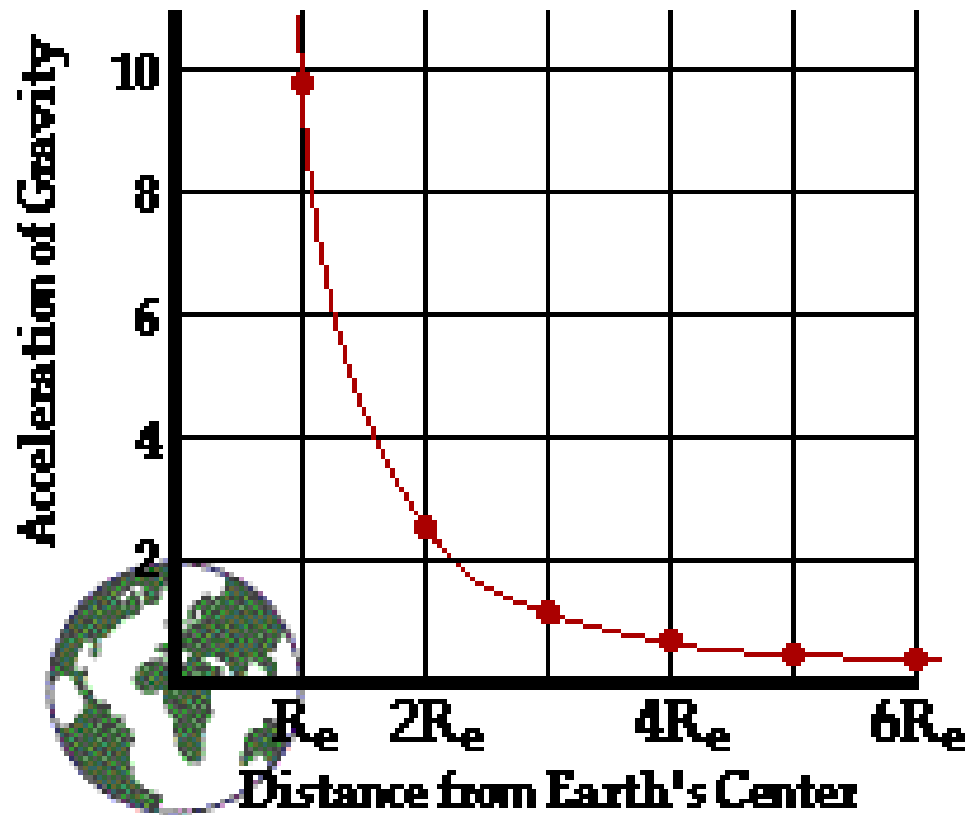
Since the force acts over a distance, it is known as a **field force**.

This is an example of an *inverse square law* of force.



Change of gravitational force with distance

The change of gravitational force with distance follows the *inverse square law*.



Practice 7

Newton's Law of Gravitation

Two spheres of mass 35 kg are 60 m apart.

- a) What force does one exert on the other?**
- b) If the mass of one is tripled and the distance is quadrupled, how does the force change?**

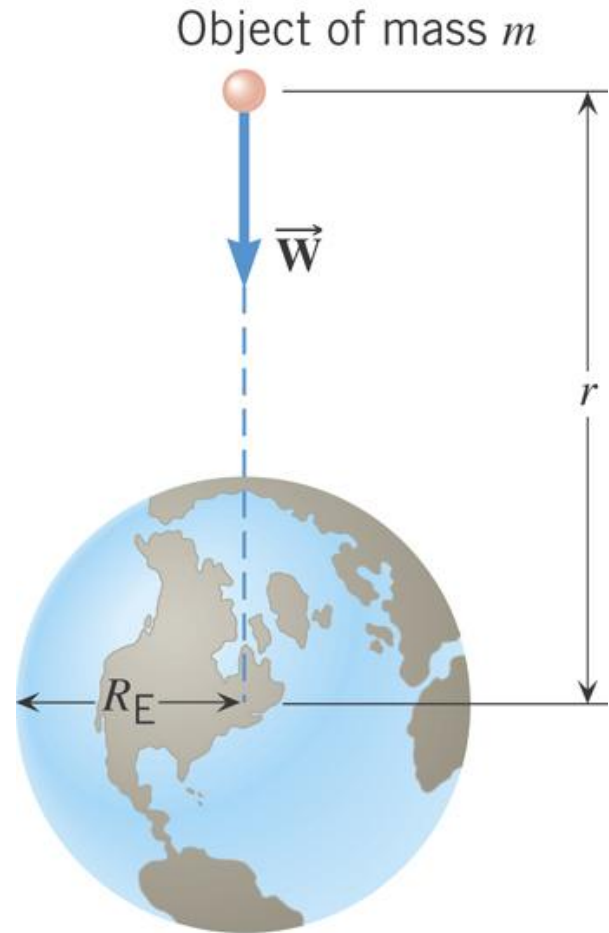
Ans.: a) $F = 3.9 \times 10^{-11} \text{ N}$; b) $F' = \frac{3}{16} F$

Gravitational force, weight & gravity acceleration

$$F_G = G \frac{M_E m}{r^2}$$

$$F_G = w = mg$$

$$g = G \frac{M_E}{r^2}$$



Mass of earth = M_E

5-7 Gravity Near the Earth's Surface; Geophysical Applications

What we refer to as the object's weight is actually the gravitational force that acts on the object (due to the earth):

$$mg = G \frac{m \times m_E}{r_E^2}$$

Hence, solving for gravity acceleration, g :

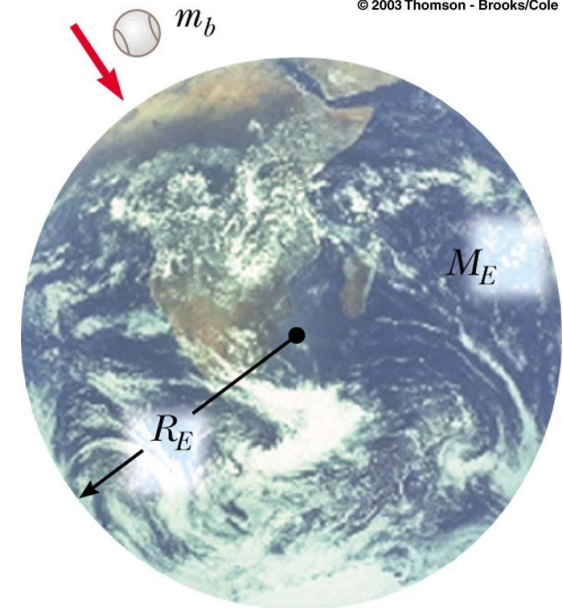
$$g = G \frac{m_E}{r_E^2}$$

5-7 Gravity Near the Earth's Surface; Geophysical Applications

Mass of the earth

After G was determined, Cavendish was able to calculate the mass of the earth (using $g = 9.80 \text{ m/s}^2$ and $R_E = 6.38 \times 10^6 \text{ m}$):

$$m_E = \frac{gr_E^2}{G} = \frac{(9.80 \text{ m/s}^2)(6.38 \times 10^6 \text{ m})^2}{6.67 \times 10^{-11} \text{ N} \cdot \text{m}^2/\text{kg}^2} = 5.98 \times 10^{24} \text{ kg}$$





INNER SPACE



Down in a cave below the surface of the earth there is:

- a) More gravity than at the earth's surface
- b) Less gravity than at the earth's surface
- c) The same gravity as at the surface



Practice 8**Newton's Law of Gravitation**

Derive the formula for gravity acceleration g at the moon's surface.

Thus, calculate the value of g on the moon given the mass of the moon = 7.35×10^{22} kg and its radius = 1.74×10^6 m.

Ans.: $g_{\text{moon}} = 1.62 \text{ m/s}^2$

5-7 Gravity Near the Earth's Surface; Geophysical Applications

Altitude, local geology, and the shape of the Earth (which is not quite spherical) influence the gravity acceleration.

**TABLE 5–1
Acceleration Due to Gravity
at Various Locations on Earth**

Location	Elevation (m)	g (m/s²)
New York	0	9.803
San Francisco	0	9.800
Denver	1650	9.796
Pikes Peak	4300	9.789
Sydney, Australia	0	9.798
Equator	0	9.780
North Pole (calculated)	0	9.832

Newton's Law of Universal Gravitation revisited..

1. The moon and the Earth are attracted to each other by gravitational force. Does the more massive earth attract the moon with a greater force than the moon attracts Earth? Explain.
2. What happens to the gravitational force between two masses when the distance between the masses is doubled?
3. If the Earth were twice as massive but remained the same size, what would happen to the value of G and g ?
4. Jupiter has about 300 times the mass of Earth and about 10 times earth's radius. Estimate the size of g on the surface of Jupiter.

Free-Fall Acceleration and the Gravitational Force

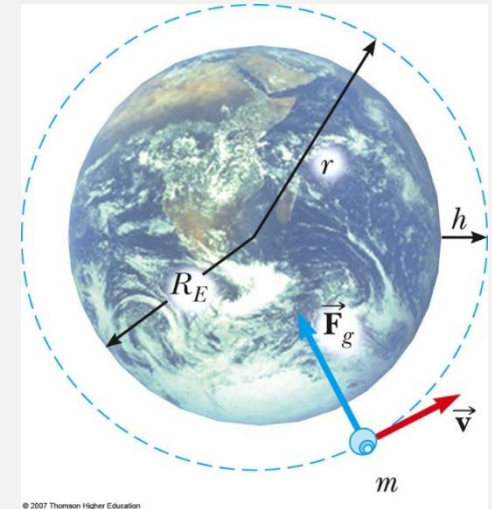
- Consider an object of mass m near the Earth's surface

$$F = G \frac{m_1 m_2}{r^2} = G \frac{m M_E}{R_E^2}$$

- Acceleration a_g due to gravity

$$F = G \frac{m M_E}{R_E^2} = m a_g$$

- Since $M_E = 5.9742 \times 10^{23}$ kg and $R_E = 6378.1$ km ,
we find at the Earth's surface $a_g = G \frac{M_E}{R_E^2} = 9.8 \text{ m/s}^2$



Free-fall acceleration, g

- Have you heard this claim:
Astronauts are weightless in space, therefore there is no gravity in space. **TRUE / FALSE**
- It is true that if an astronaut on the International Space Station (ISS) tries to step on a scale, he/she will weigh nothing. **TRUE / FALSE**
- It may seem reasonable to think that if $\text{weight} = mg$: since $\text{weight} = 0$, $g = 0$. **TRUE / FALSE**
- If you stand on a scale in an elevator and then the cables are cut, you will also weigh nothing. **TRUE / FALSE**
- Astronauts in orbit are in free-fall around the Earth, just as you would be in the elevator. They do not fall to Earth only due to their very high tangential speed. **TRUE / FALSE**

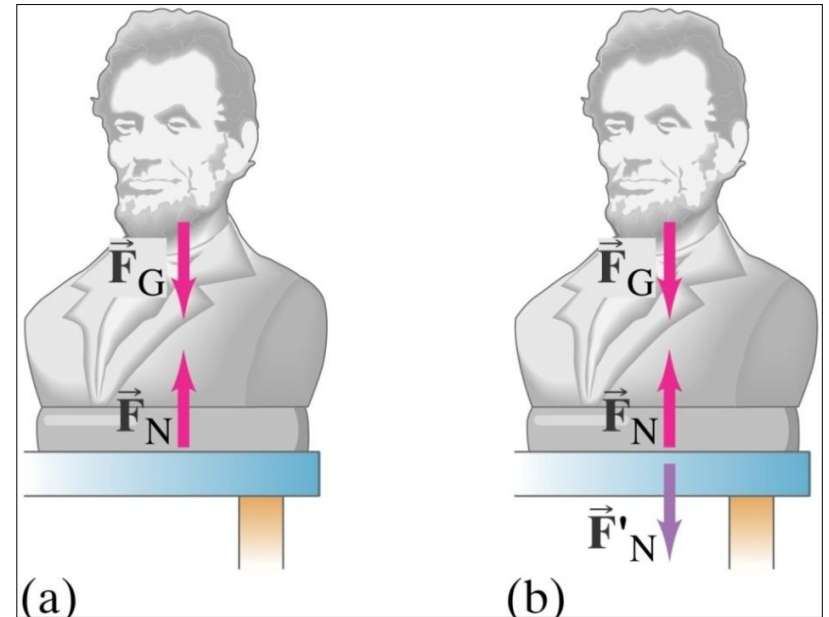
4-3 Weight – the Force of Gravity; the Normal Force

According to Newton's laws, an object at rest must have no net force on it, i.e. $\Sigma \vec{F} = 0$.

A statue is sitting on a table, the force of gravity is still there, thus what other force is acting?

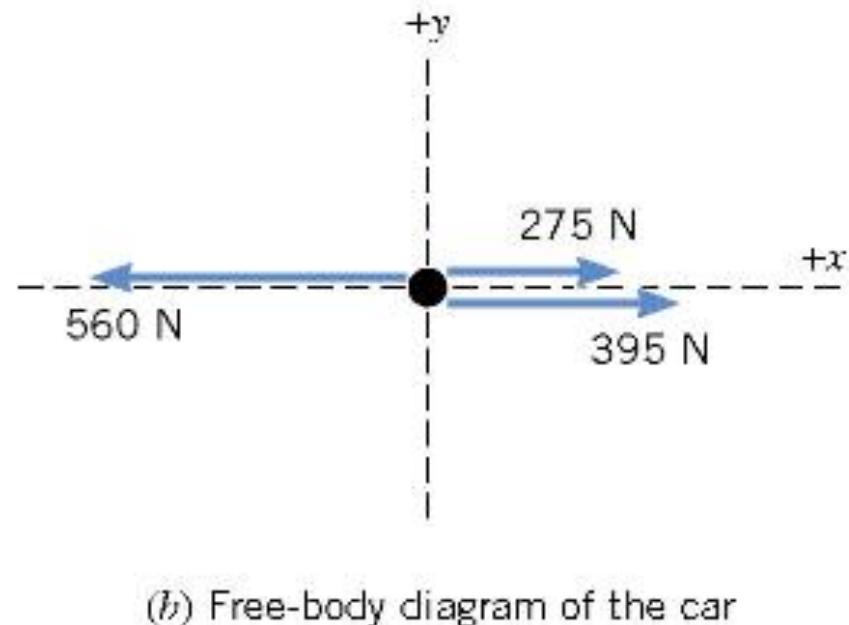
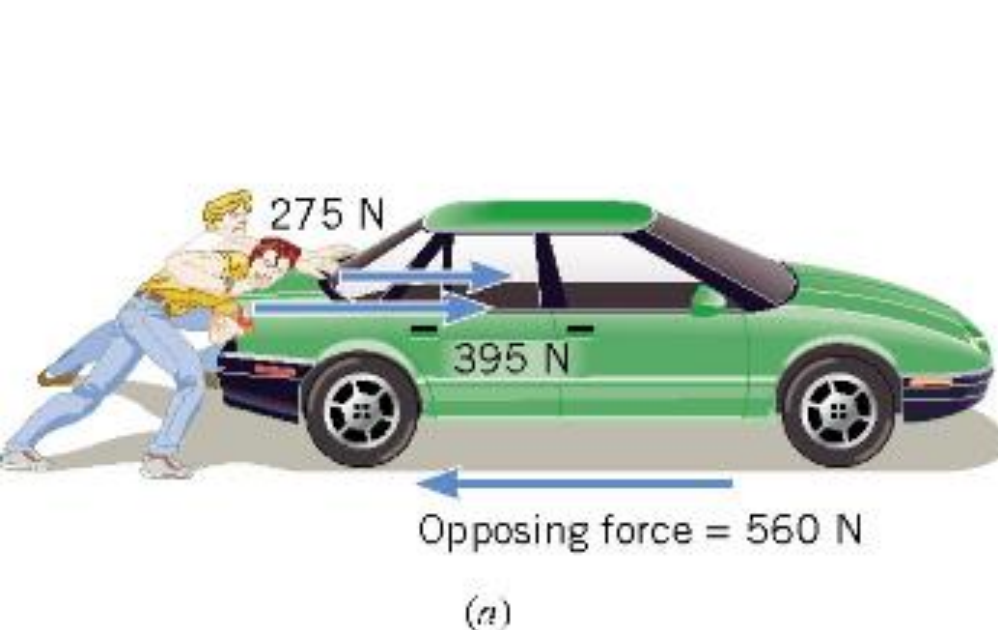
It is a force exerted perpendicular to the surface, called **normal force**.

In order to balance the gravity force, its magnitude must be the same as the gravity force. (if the required force gets too big, something breaks!)



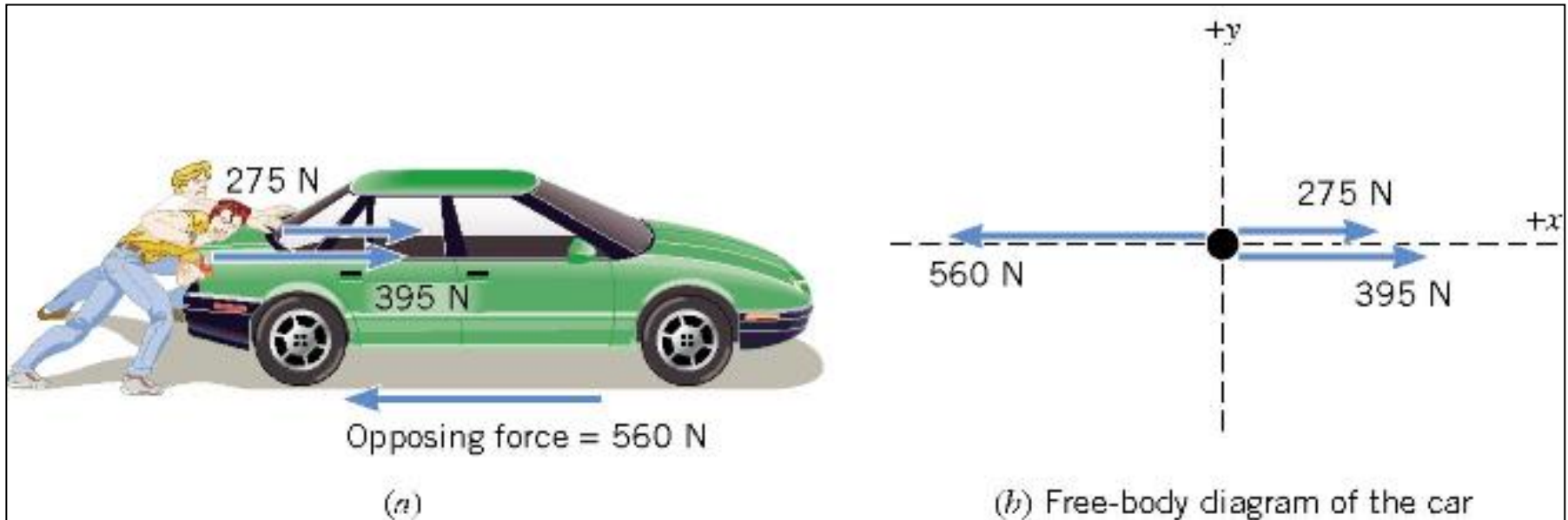
4-4 Solving Problems with Newton's Laws – Free-Body Diagrams

A ***free-body-diagram*** is a diagram that represents the object and the forces that act on it.



Newton's Second Law of Motion

Example 2



The net force in this case is:

$$275 \text{ N} + 395 \text{ N} - 560 \text{ N} = +110 \text{ N}$$

and is directed along the $+x$ axis of the coordinate system.

Answer

If the mass of the car is 1850 kg then, by Newton's second law, the acceleration is

$$a = \frac{\sum F}{m} = \frac{+110 \text{ N}}{1850 \text{ kg}} = +0.059 \text{ m/s}^2$$

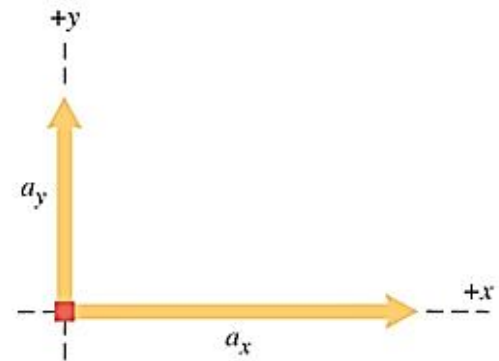
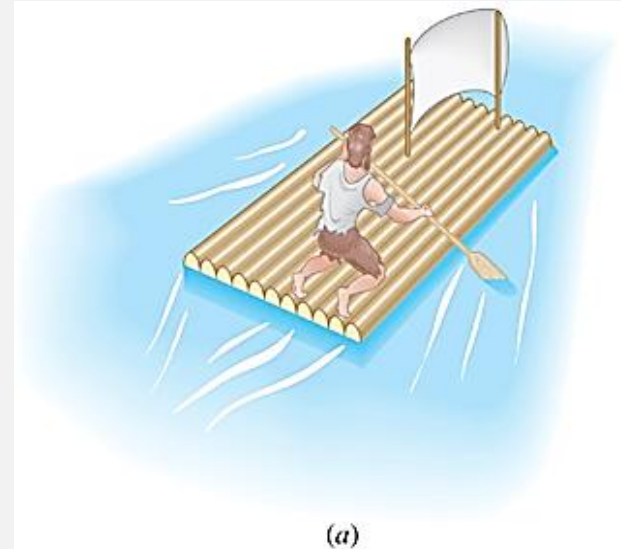
Example 3

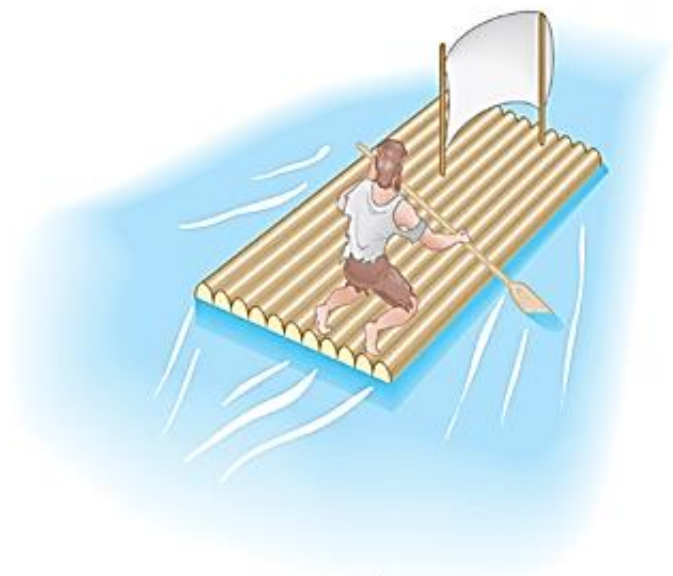
A boy is stranded on a raft (mass of the boy and raft = 1300 kg).

By paddling, he exerts a force of 17 N due east.

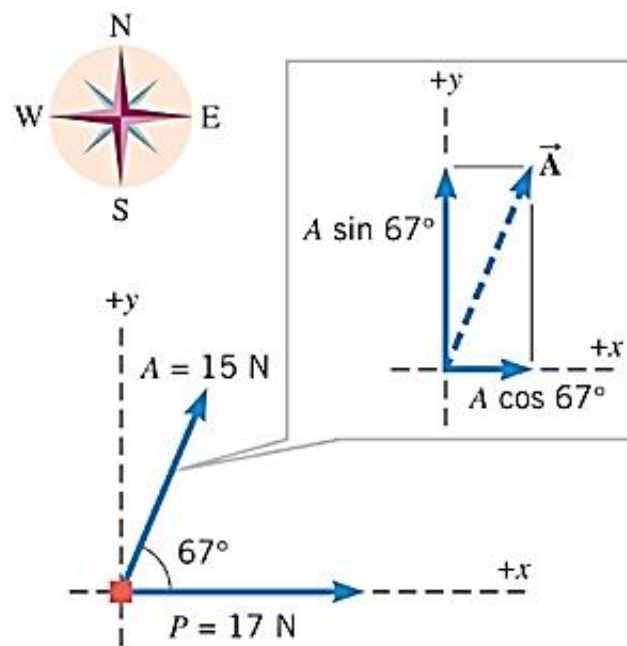
The wind exerts a force of 15 N directed 67° north of east on the raft.

Determine the x - and y -component of the raft's acceleration.



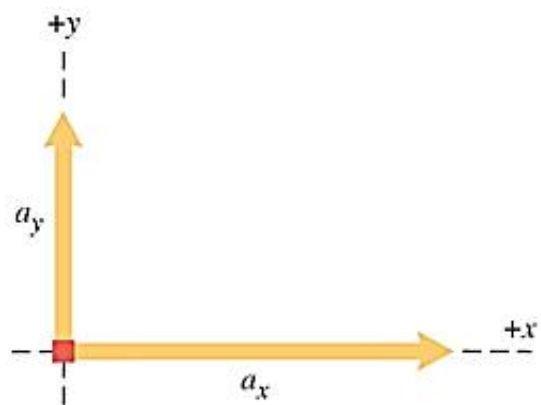


(a)

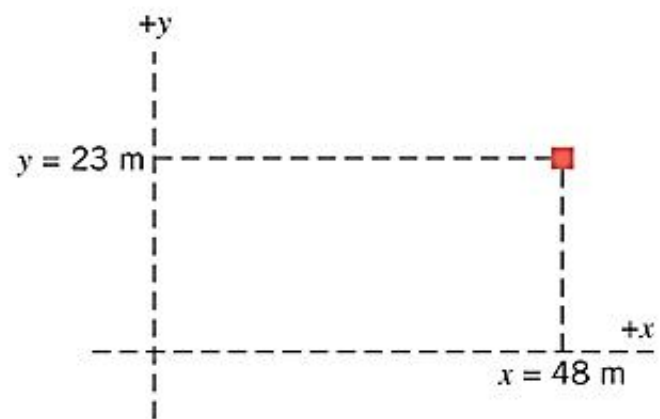


Free-body diagram of the raft

(b)



(c)



(d)

The Vector Nature of Newton's Second Law

The net force on the raft can be calculated in the following way:

Force	<i>x</i>-component	<i>y</i>-component
$\vec{P}$	+17 N	0 N
$\vec{A}$	$+(15 \text{ N}) \cos 67$	$+(15 \text{ N}) \sin 67$
	+23 N	+14 N

We can continue to analyze separately in x - and y - components:

$$a_x = \frac{\sum F_x}{m} = \frac{+23 \text{ N}}{1300 \text{ kg}} = +0.018 \text{ m/s}^2$$

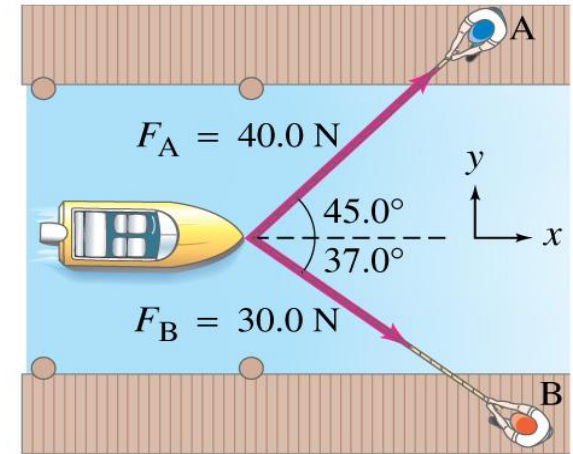
$$a_y = \frac{\sum F_y}{m} = \frac{+14 \text{ N}}{1300 \text{ kg}} = +0.011 \text{ m/s}^2$$

Students task:

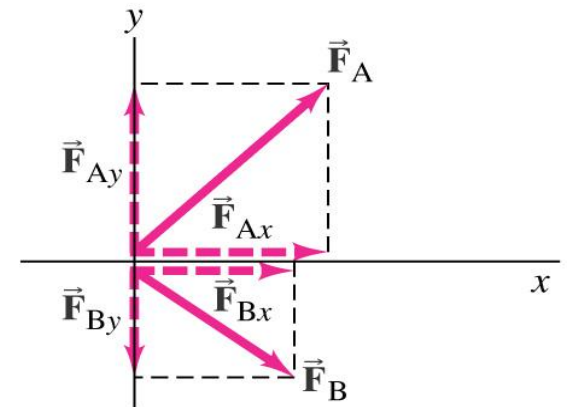
What is then the magnitude and direction of the net acceleration of the raft.

4-4 Solving Problems with Newton's Laws – Free-Body Diagrams

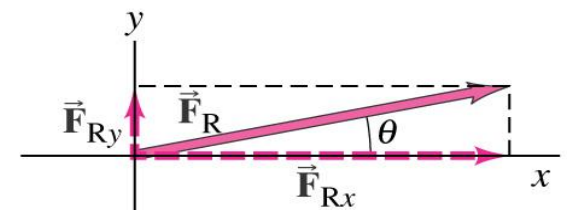
1. Draw a sketch.
2. For one object, draw a free-body diagram, showing all the forces acting ON the object. Make the magnitudes and directions as accurate as you can. Label each force. If there are multiple objects, draw a separate diagram for each one.
3. Resolve vectors into components.
4. Apply Newton's second law to each component.
5. Solve.



(a)



(b)



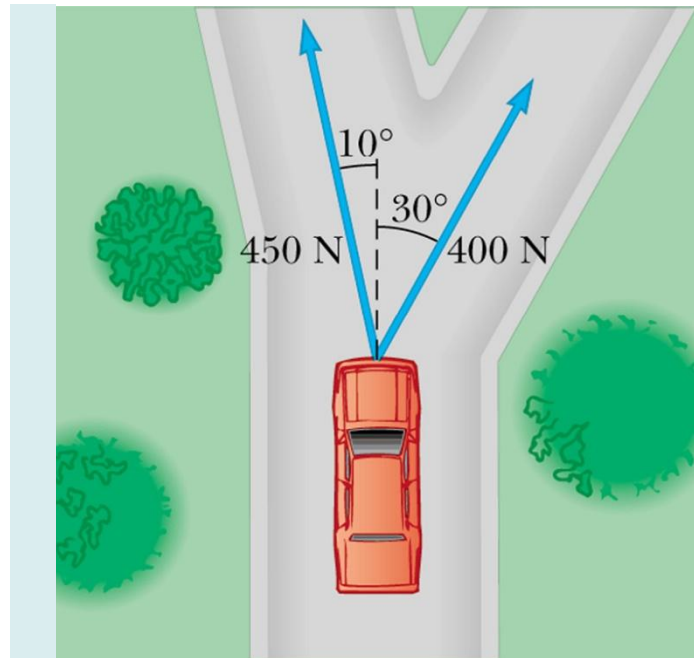
(c)

Practice Questions (I)

- a) A freight train has a mass of 1.5×10^7 kg. If the locomotive can exert a constant pull of 7.5×10^5 N, how long does it take to increase the speed of the train from rest to 80 km/h?
- b) A bag of sugar weighs 5.00 lb on Earth. What should it weigh in newtons on the Moon, where the free-fall acceleration is $1/6$ that on Earth? Repeat for Jupiter, where g is 2.64 times that on Earth. Find the mass of the bag of sugar in kilograms at each of the three locations.
[1 lb $\approx$ 4.45 N]
- c) A performer in a circus is fired from a cannon as a “human cannonball” and leaves the cannon with a speed of 18.0 m/s. The performer’s mass is 80.0 kg. The cannon barrel is 9.20 m long. Find the average net force exerted on the performer while he is being accelerated inside the cannon.

Practice Questions (I)

- d) Two forces are applied to a car in an effort to move it.
- What is the resultant of these two forces?
 - If the car has a mass of 3000 kg, what acceleration does it have? Ignore friction.



4-5 Weight and Normal Force on a Box (I)

(a)

$$\Sigma F_y = F_N - mg = 0$$

$$F_N = mg = (10.0 \text{ kg})(9.8 \text{ m.s}^{-2}) \\ = 98.0 \text{ N}$$

(b)

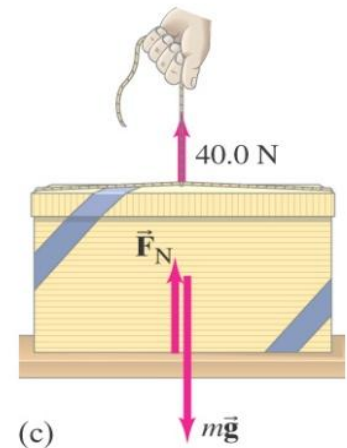
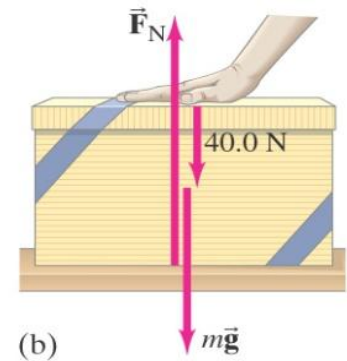
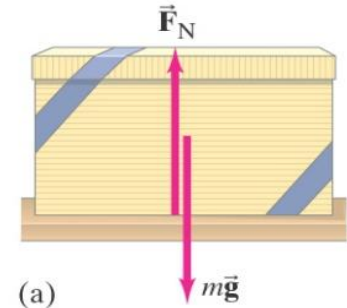
$$\Sigma F_y = F_N - mg - 40.0 \text{ N} = 0$$

$$F_N = mg + 40.0 \text{ N} = 138.0 \text{ N}$$

(c)

$$\Sigma F_y = F_N - mg + 40.0 \text{ N} = 0$$

$$F_N = mg - 40.0 \text{ N} = 58.0 \text{ N}$$



4-5 Weight and Normal Force on a Box (II)

Accelerating the Box

What if $F_P > mg$, $F_P = 100 \text{ N}$?

$$\begin{aligned}\Sigma F_y &= F_P + F_N - mg \\ &= 100 \text{ N} + F_N - 98.0 \text{ N}\end{aligned}$$

Let $\Sigma F_y = 0$,

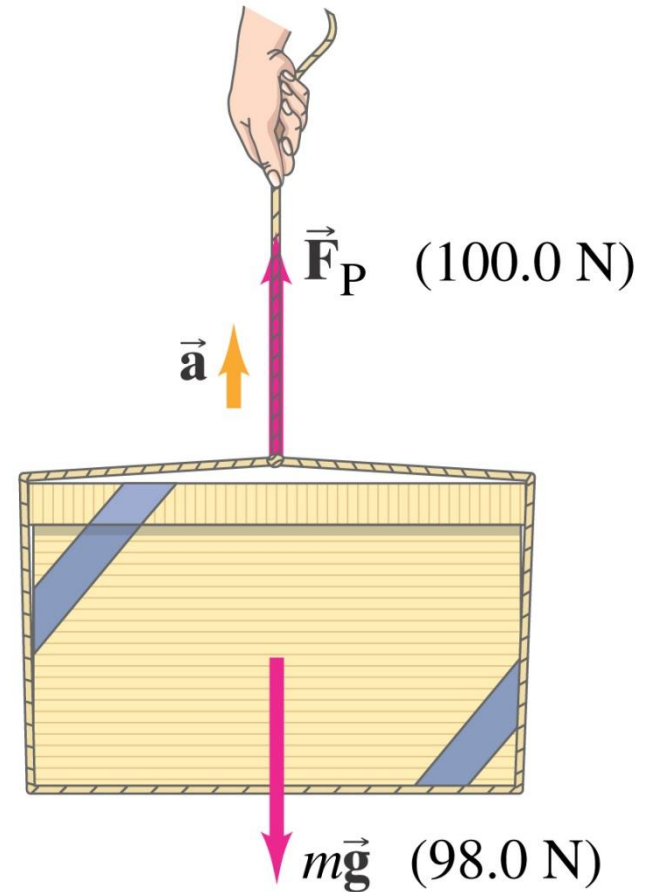
$$\begin{aligned}0 &= 100 \text{ N} + F_N - 98.0 \text{ N}, \\ F_N &= -2.0 \text{ N} \text{ (unreasonable!)}\end{aligned}$$

Therefore $F_N = 0$, and

$$\Sigma F_y = ma = 100 \text{ N} - 98.0 \text{ N} = 2.0 \text{ N}$$

Acceleration of the box

$$a = \Sigma F_y / m = 2.0 \text{ N} / 10.0 \text{ kg} = 0.20 \text{ m/s}^2$$



4-5 Pulling a box at an angle

Find normal force, F_N and acceleration, a .

- Resolve the pulling force into x - and y -components:

$$F_{Px} = 40.0 \text{ N} \cos 30.0^\circ = 34.6 \text{ N}$$

$$F_{Py} = 40.0 \text{ N} \sin 30.0^\circ = 20.0 \text{ N}$$

- Apply Newton's 2nd law,

in y -direction:

$$\Sigma F_y = F_N + F_{Py} - mg = 0$$

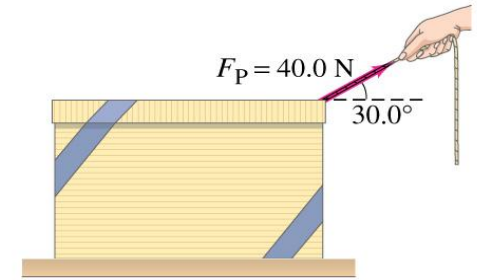
$$F_N = mg - F_{Py} = (10)(9.80) - 20.0 \text{ N} = 78.0 \text{ N}$$

in x -direction:

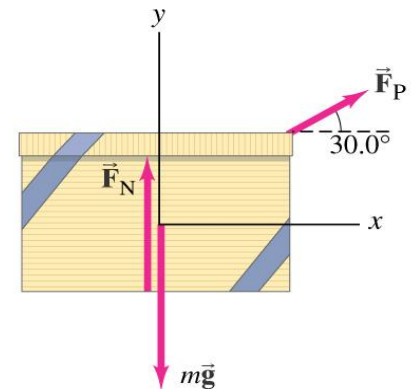
$$\Sigma F_x = ma = F_{Px} = 34.6 \text{ N}$$

Acceleration of the box

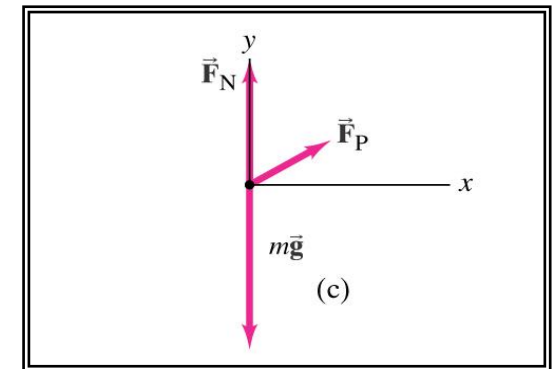
$$\begin{aligned} a_x &= \Sigma F_x / m = 34.6 \text{ N} / 10 \text{ kg} \\ &= 3.46 \text{ m.s}^{-2} \end{aligned}$$



(a)



(b)



(c) Free-body diagram

4-5 Solving Problems with Newton's Laws

Find the acceleration, a and the tension force in the cord, F_T :

► Boxes A and B are connected by a lightweight cord on a smooth frictionless surface. Horizontal force $F_P = 40.0$ N is applied to box A.

► In x -direction,

Box A: $\Sigma F_x = F_P - F_T = m_A a \quad \cdots \text{(i)}$

Box B: $\Sigma F_x = F_T = m_B a \quad \cdots \text{(ii)}$

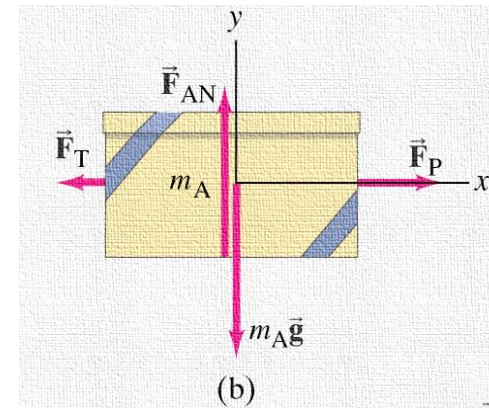
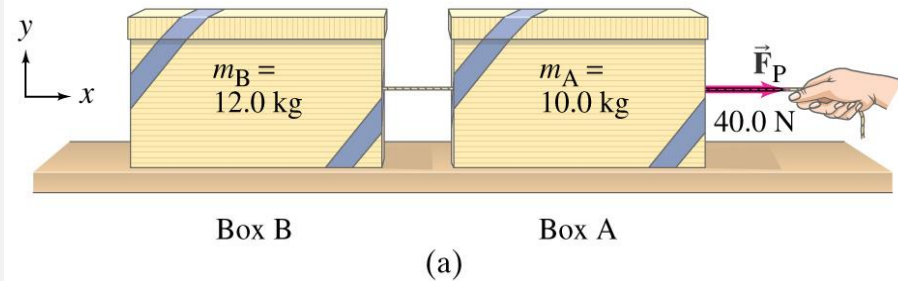
To obtain a :

$$\text{(i)+(ii)} \Rightarrow F_P = (m_B + m_A)a$$

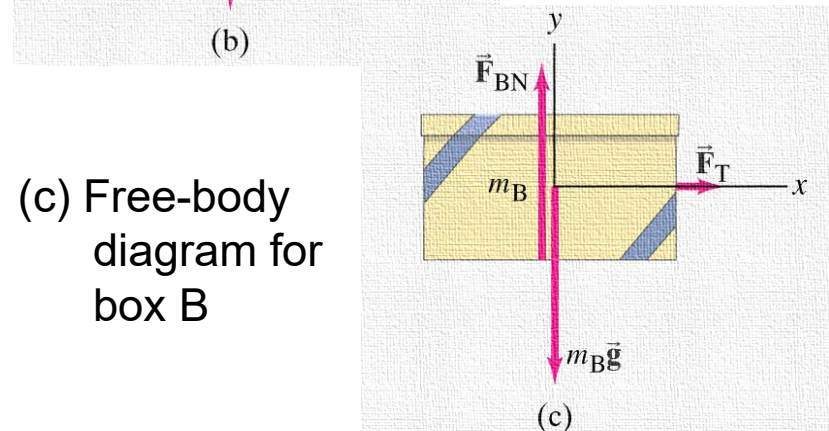
$$40.0 = (12.0 + 10.0)a$$

$$a = 1.25 \text{ m.s}^{-2}$$

$$\text{(ii)} \Rightarrow F_T = m_B a = (12.0)(1.25) = 15.0 \text{ N}$$



(b) Free-body diagram for box A

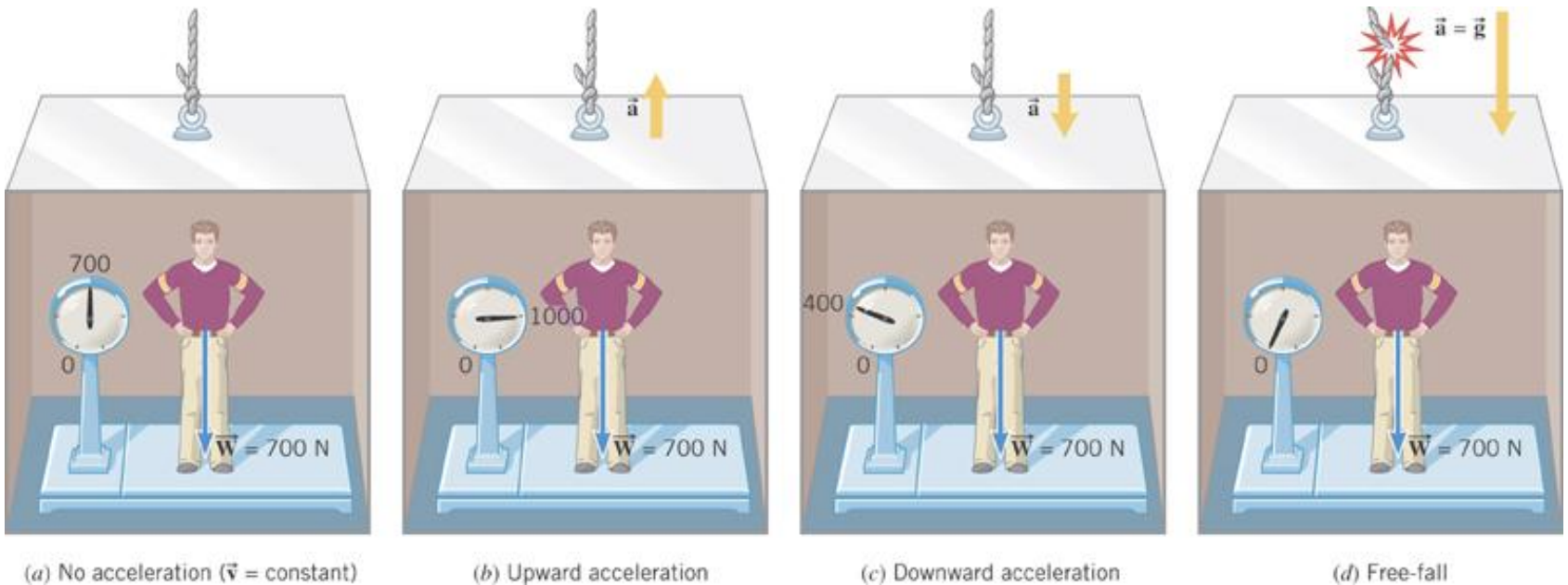


(c) Free-body diagram for box B

Apparent Weight

The **apparent weight** of an object is the reading of the scale.

It is equal to the normal force the man exerts on the scale.



Apparent weight

A 65 kg woman descends in an elevator that briefly accelerates at $a = 0.20g$ downwards.

- (a) During this time, what is her weight?
What does the scale read?

Weight, $w = mg = 65 \times 9.80 = 637 \text{ N}$

Scale reading, $F_N = ?$

$$F_{\text{net}} = \sum F = ma$$

$$F_N - mg = ma \quad \Rightarrow \quad F_N - mg = m(-0.20g)$$

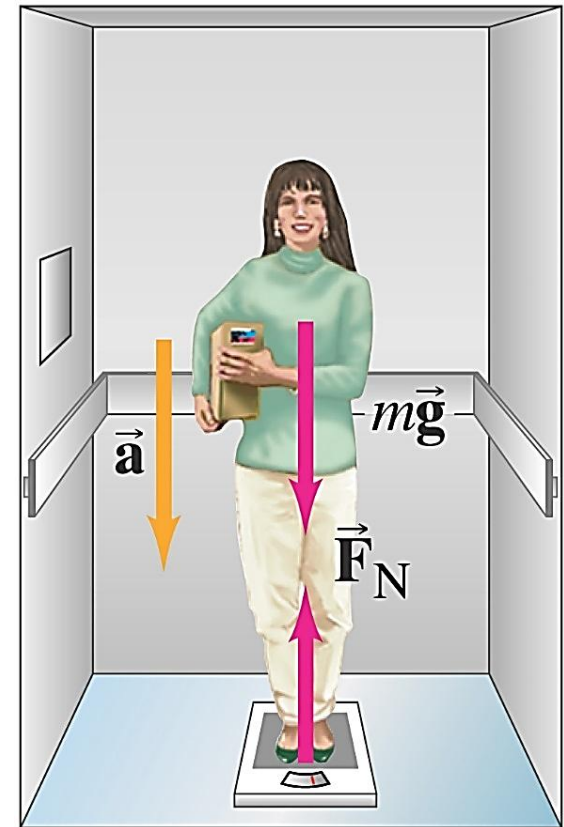
$$F_N = m(g - 0.20g) = 65(0.80g) = 510 \text{ N}$$

- (b) What does the scale read when the elevator descends at constant speed 2.0 m/s?

$$F_{\text{net}} = \sum F = ma = 0$$

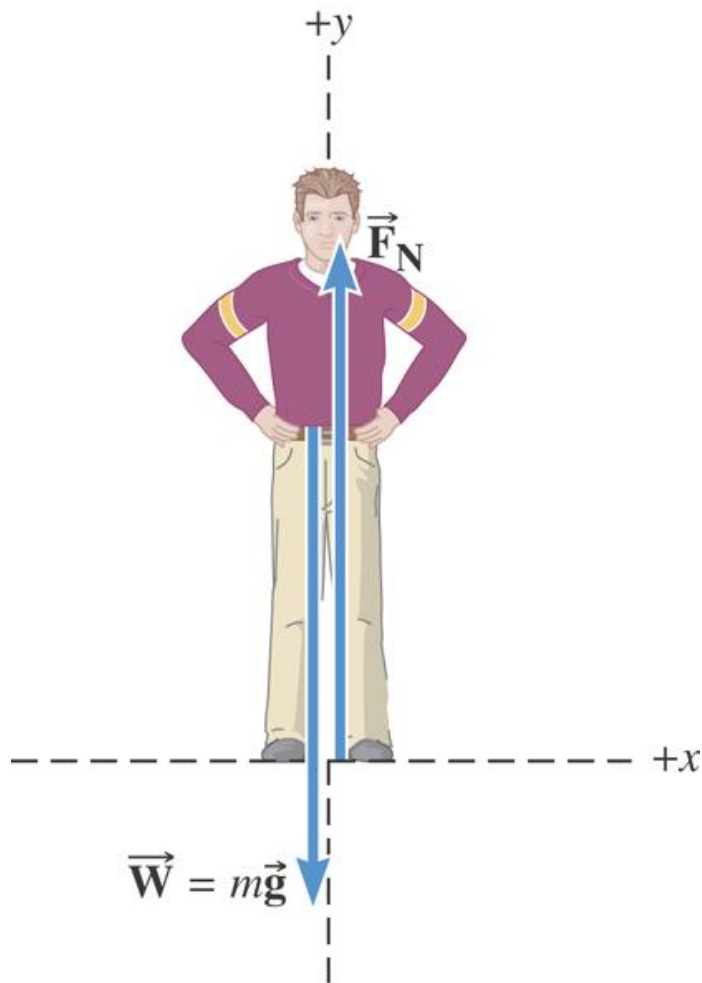
$$F_N - mg = 0$$

$$F_N = mg = 637 \text{ N}$$



Apparent weight

The normal force



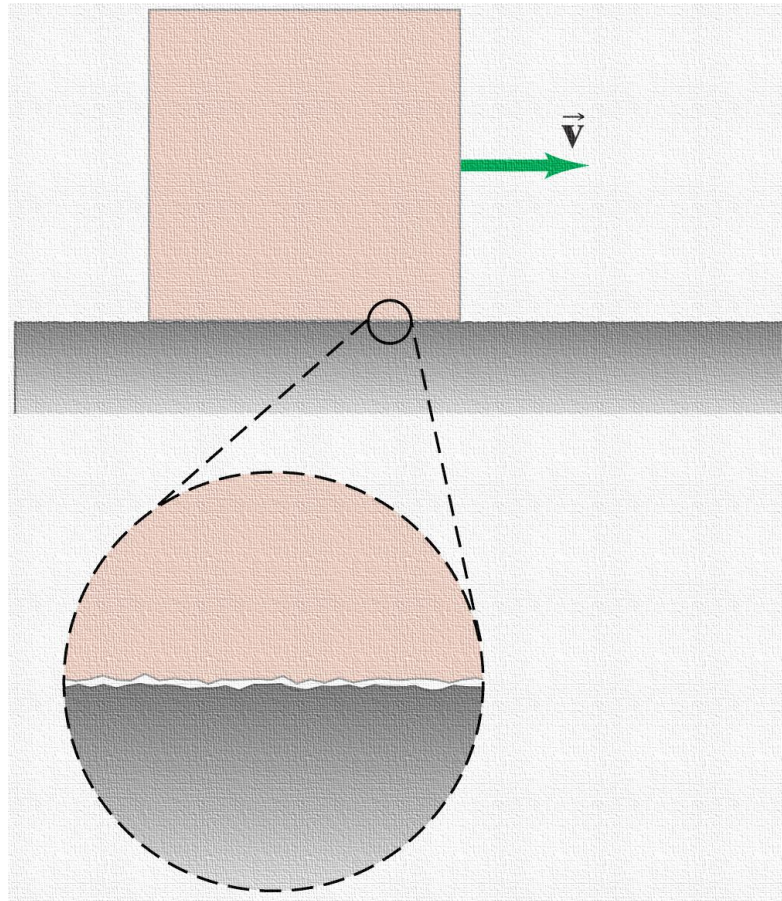
$$\sum F_y = +F_N - mg = ma$$

 **apparent weight**

 **true weight**

4-6 Friction Force

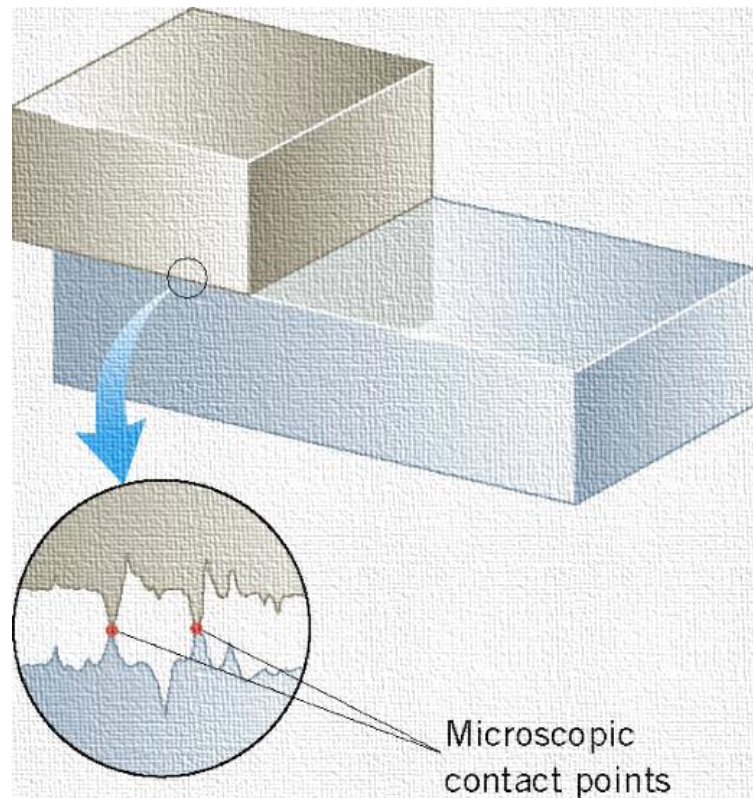
On a microscopic scale, most surfaces are rough.



4-6 Friction Force

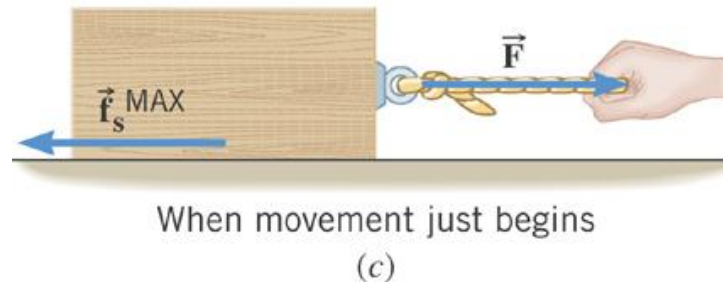
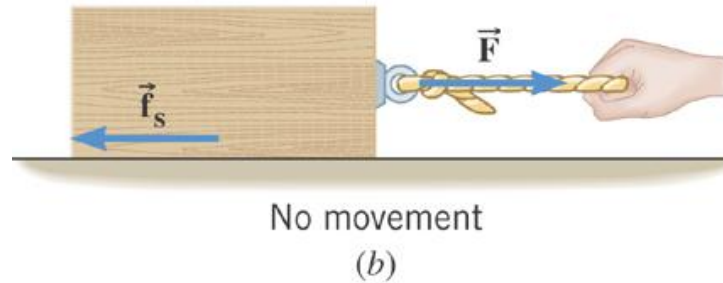
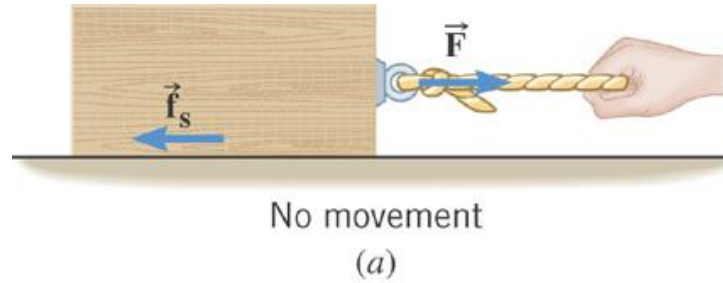
Static Frictional Force

When an object is in contact with a surface there is a force acting on that object. The component of this force that is parallel to the surface is called the **frictional force**.



Static Frictional Force

When the two surfaces are not sliding across one another the friction is called **static friction**.



4-6 Friction Force

STATIC FRICTION

A force is applied on a heavy object at rest. The object does not begin to move yet. During this time, static friction $F_{\text{fr,s}}$ acts to oppose the applied force F_{app} where:

$$F_{\text{fr,s}} = F_{\text{app}}$$

The moment the applied force exceeds the **maximum static friction force** ($F_{\text{fr, max}}$), it starts to move, where:

$$F_{\text{fr, max}} = \mu_s F_N$$

where μ_s = **coefficient of static friction**, where $0 < \mu_s < 1$
(which depends on the type and nature of the two surfaces in contact)

F_N = normal force

4-6 Friction Force

STATIC FRICTION

During the process when the force is applied but the object remains static, static friction force varies from zero to maximum static friction,

$$0 \leq F_{\text{fr},s} \leq F_{\text{fr},s,\text{max}}$$

4-6 Friction Force

KINETIC FRICTION

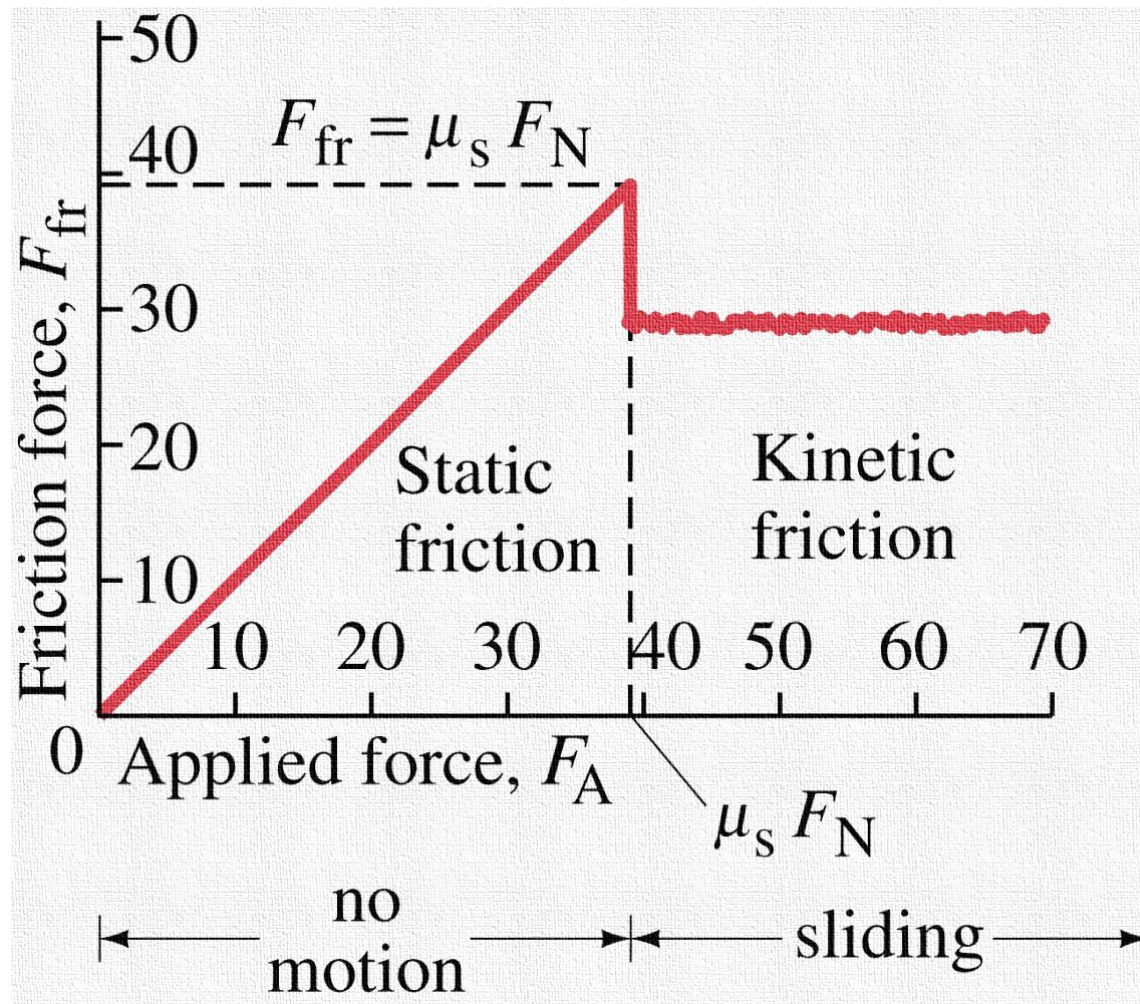
- The moment the object begins to slide, static friction is replaced by **kinetic friction force**.
- Kinetic friction force is given by:

$$F_{\text{fr}} = \mu_k F_N$$

where μ_k = coefficient of kinetic friction,
given between $0 < \mu_k < 1$

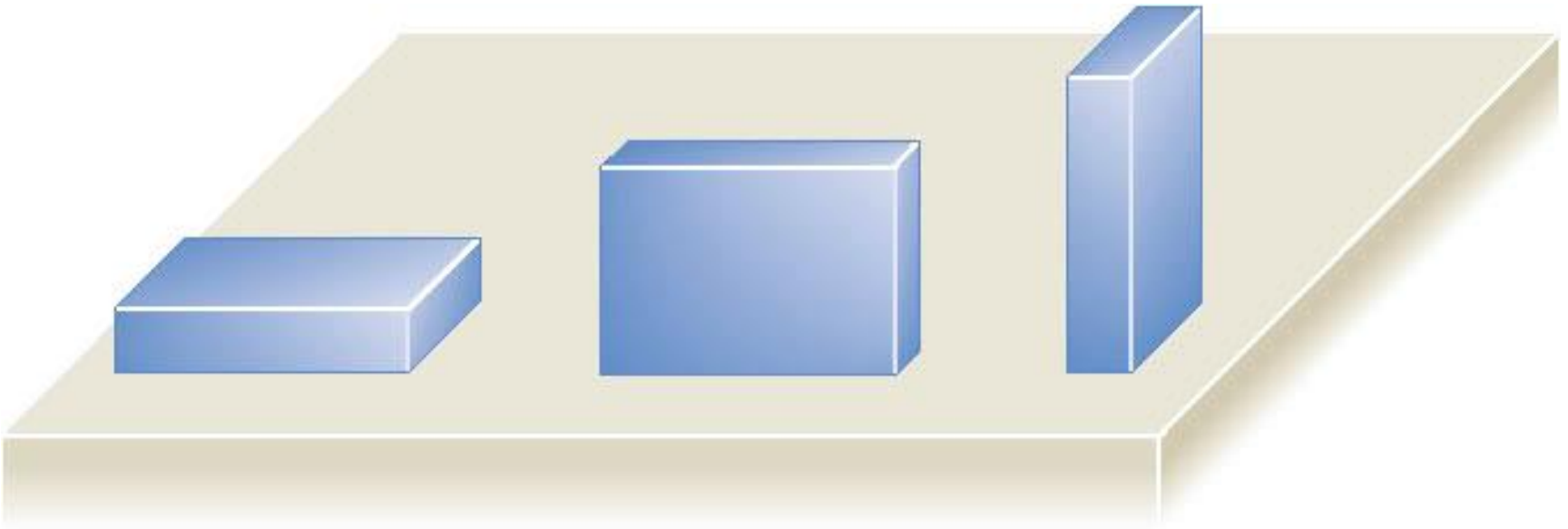
4-6 Friction force

The plot shows how the friction force varies with the applied force.



4-6 Friction Force

The magnitude of the frictional force does not depend on the contact area of the surfaces.



4-6 Friction Force

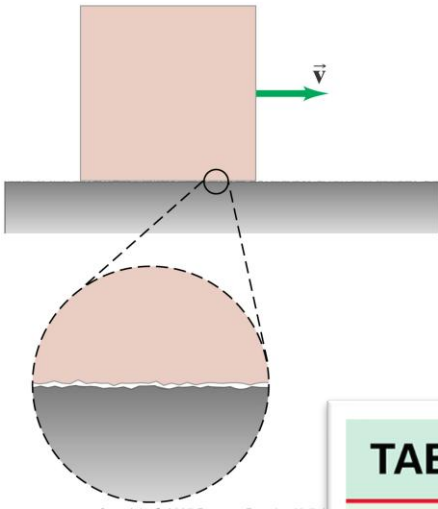


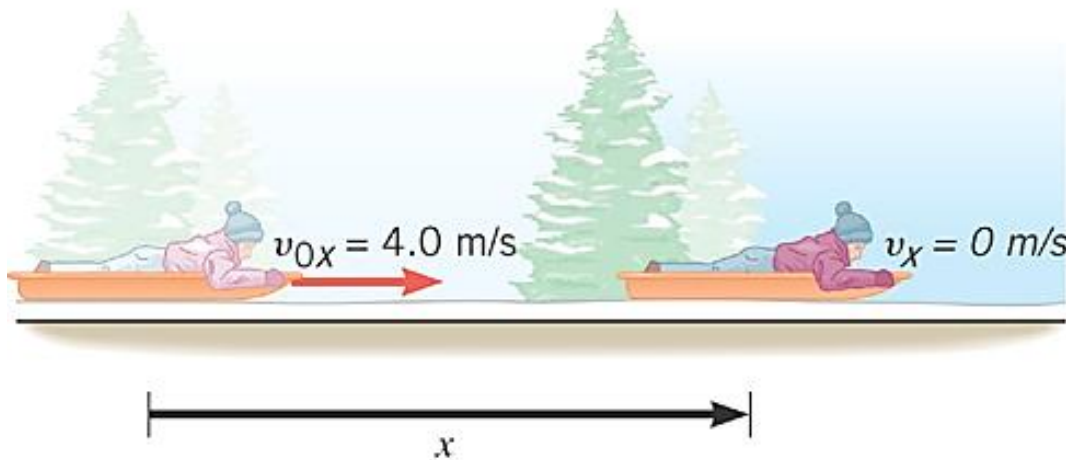
TABLE 4–2 Coefficients of Friction[†]

Surfaces	Coefficient of Static Friction, μ_s	Coefficient of Kinetic Friction, μ_k
Wood on wood	0.4	0.2
Ice on ice	0.1	0.03
Metal on metal (lubricated)	0.15	0.07
Steel on steel (unlubricated)	0.7	0.6
Rubber on dry concrete	1.0	0.8
Rubber on wet concrete	0.7	0.5
Rubber on other solid surfaces	1–4	1
Teflon [®] on Teflon in air	0.04	0.04
Teflon on steel in air	0.04	0.04
Lubricated ball bearings	<0.01	<0.01
Synovial joints (in human limbs)	0.01	0.01

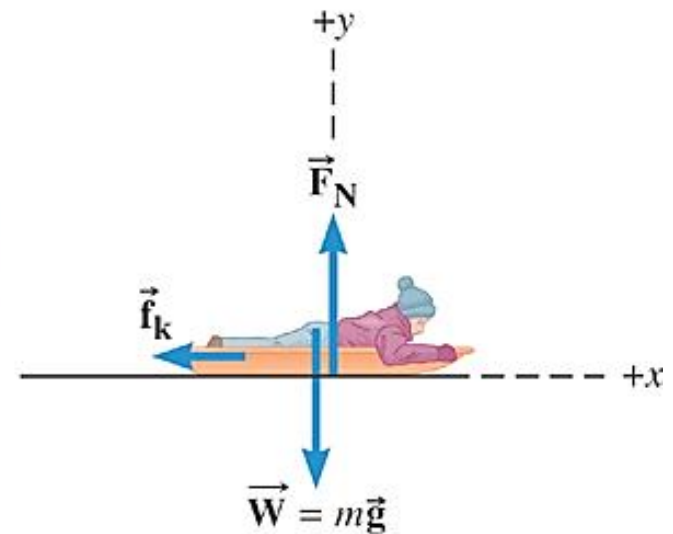
[†] Values are approximate and intended only as a guide.

PRACTICE 1:

A sled and its rider are moving at a speed of 4.0 m/s along a horizontal stretch of snow. The snow exerts a kinetic frictional force on the sled, so the sled slows down and eventually comes to a stop.



(a)



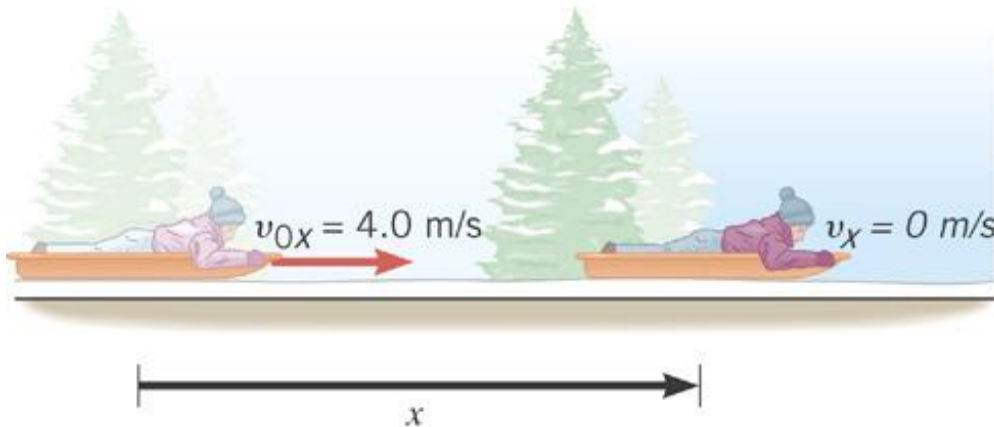
(b) Free-body diagram
for the sled and rider

Static and Kinetic Frictional Forces

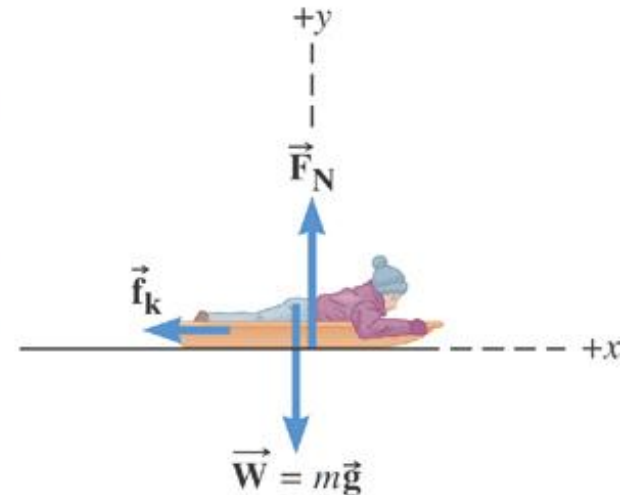
Suppose the coefficient of kinetic friction, $\mu_k = 0.05$ and the total mass (rider + sled) is 40 kg.

(a) Calculate the kinetic frictional force.

(b) How far does the sled travel before it was brought to a complete halt?

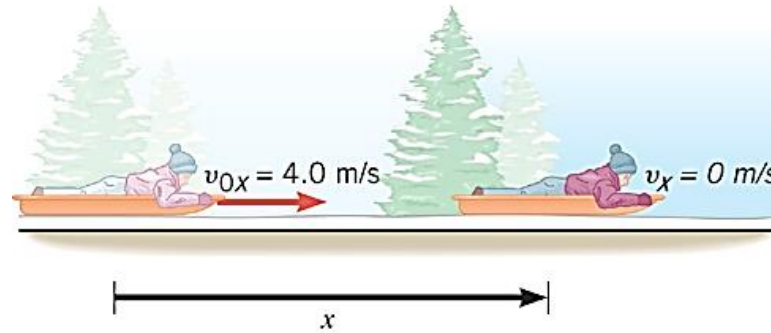


(a)

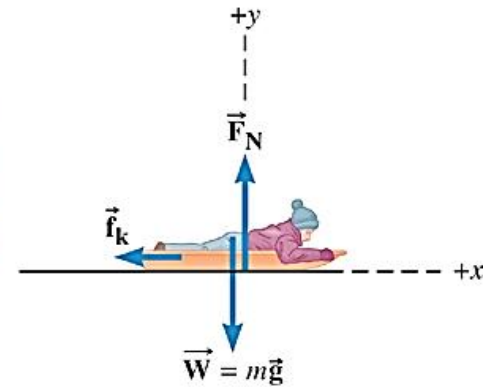


(b) Free-body diagram
for the sled and rider

Answer:



(a)



(b) Free-body diagram
for the sled and rider

(a)

$$f_k = \mu_k F_N = \mu_k mg = 0.05(40\text{kg})(9.80\text{ m/s}^2) = 20\text{N}$$

(b)

Example 4

Find acceleration of the boxes and tension force in the cord, given $\mu_k = 0.20$.

For box A ►

- In y -direction:

$$\Sigma F_y = F_N - m_A g = 0 \quad \Rightarrow F_N = m_A g$$

- In x -direction:

$$\Sigma F_x = F_T - F_{fr} = F_T - \mu_k F_N$$

$$m_A a = F_T - \mu_k m_A g \quad \dots(i)$$

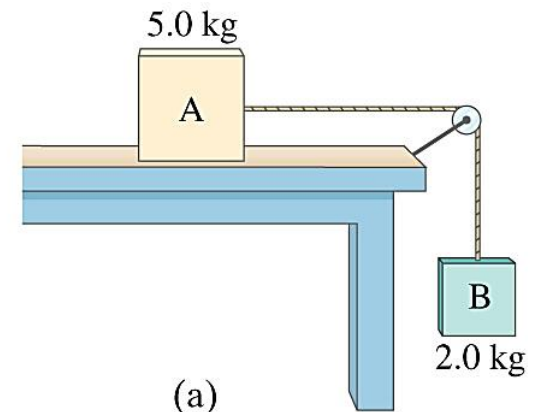
For box B ► in y -direction:

$$\Sigma F_y = m_B g - F_T$$

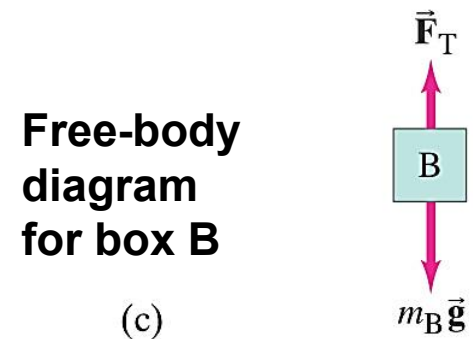
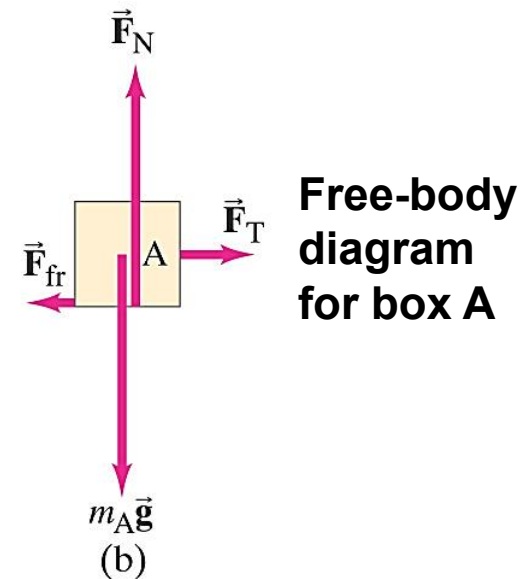
$$m_B a = m_B g - F_T \quad \dots(ii)$$

$$(i)+(ii) \Rightarrow (m_A + m_B)a = m_B g - \mu_k m_A g$$

$$a = \frac{m_B g - \mu_k m_A g}{m_A + m_B} = \frac{9.80(2.0 - 1.0)}{2.0 + 5.0} = 1.4 \text{ m/s}^2$$



(a)



Practice Questions (II)

1. A dockworker loading crates on a ship finds that a 20 kg crate, initially at rest on a horizontal surface, requires a 75 N horizontal force to set it in motion.

However, after the crate is in motion, a horizontal force of 60 N is required to keep it moving with a constant speed.

Find the

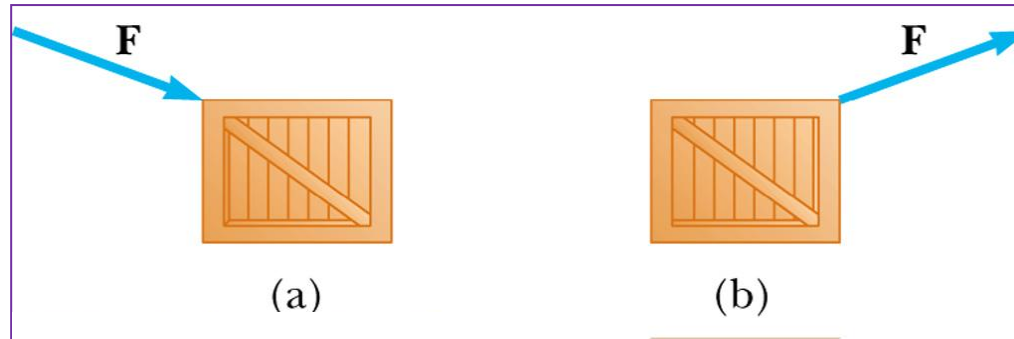
(a) coefficient of static friction μ_s , and

(b) coefficient of kinetic friction μ_k

between crate and floor.

Practice Questions (II)

2. A 1000 N crate is being pushed across a level floor at a constant speed by a force $F = 300$ N at an angle 20.0° below horizontal as shown in figure (a).



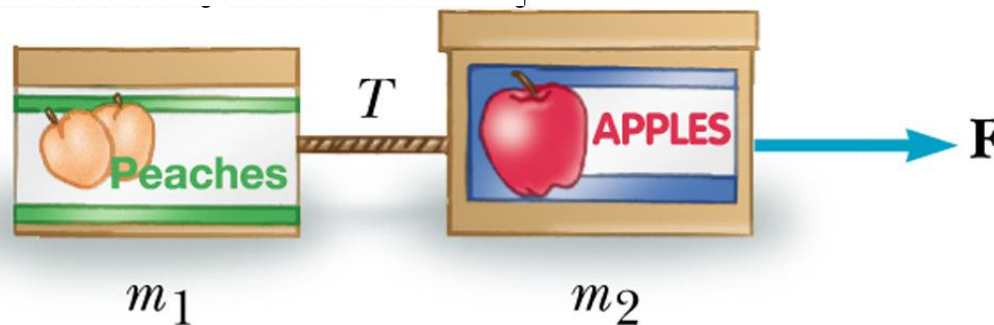
- (i) What is the normal force acting on the crate?
- (ii) Find the coefficient of kinetic friction between the crate and the floor.

The 300 N force is now pulling the block at an angle 20.0° above horizontal as shown in figure (b).

- (iii) What is the normal force acting on the crate now?
- (iv) Hence, determine the acceleration of the crate.
[Use the coefficient of friction calculated in (ii).]

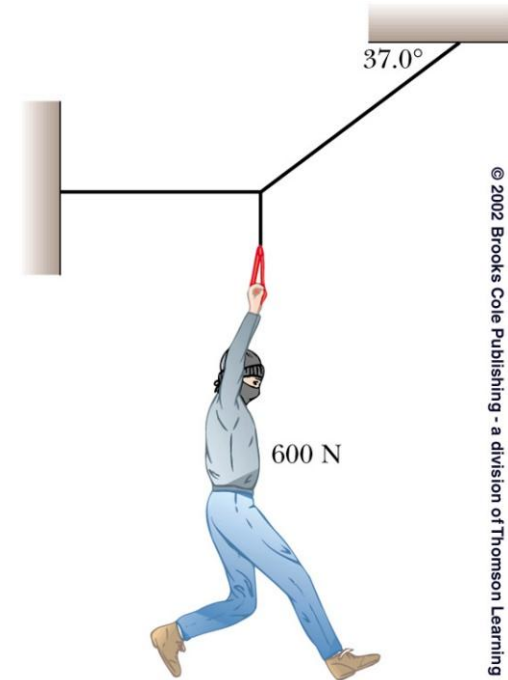
Practice Questions (II)

3. Two boxes of fruit on a frictionless horizontal surface are connected by a light string, where $m_1 = 10$ kg and $m_2 = 20$ kg. A force of 50 N is applied to the 20 kg box.
- (a) Determine the acceleration of each box and the tension in the string.
- (b) Repeat the problem for the case where there is friction and the coefficient of kinetic friction between each box and the surface is 0.10.

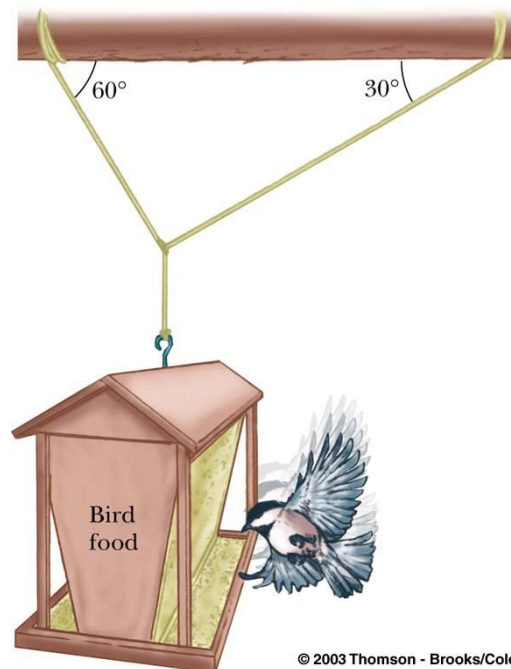


Practice Questions (II)

4. Find the tension in each cable supporting the 600 N cat burglar as shown.
5. A 150 N bird feeder is supported by three cables as shown below.
Find the tension in each cable.



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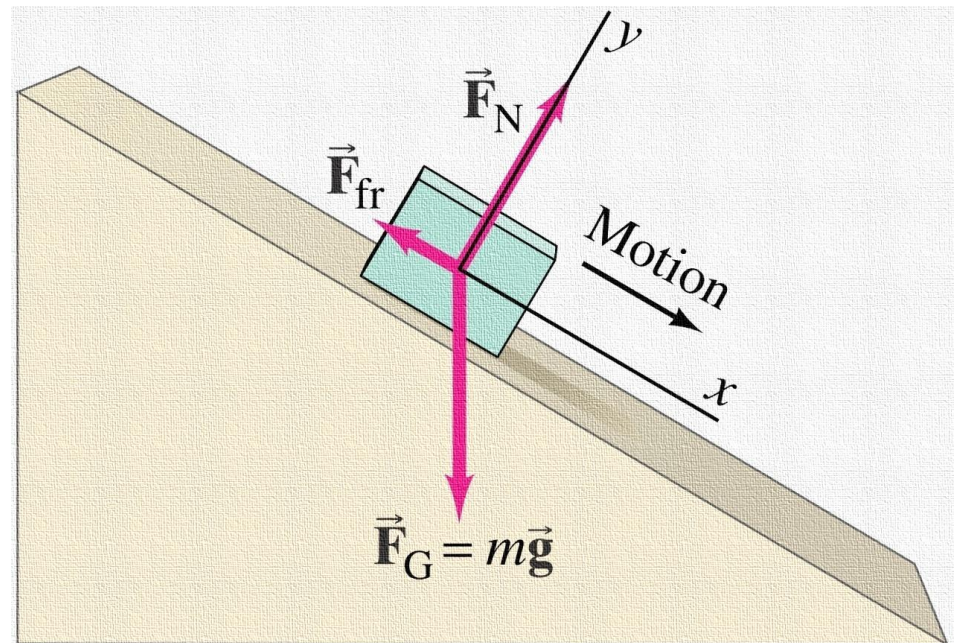
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4-6 Friction Force on Inclines

An object sliding down an incline has three forces acting: normal force, gravity force and friction force.

- The **normal force** is always perpendicular to the surface.
- The **friction force** is parallel to the surface.
- The **gravitational force** points vertically straight down.

If the object is at rest, the forces present are the same, except that we use static frictional force instead of kinetic friction. The sum of the forces then is zero.

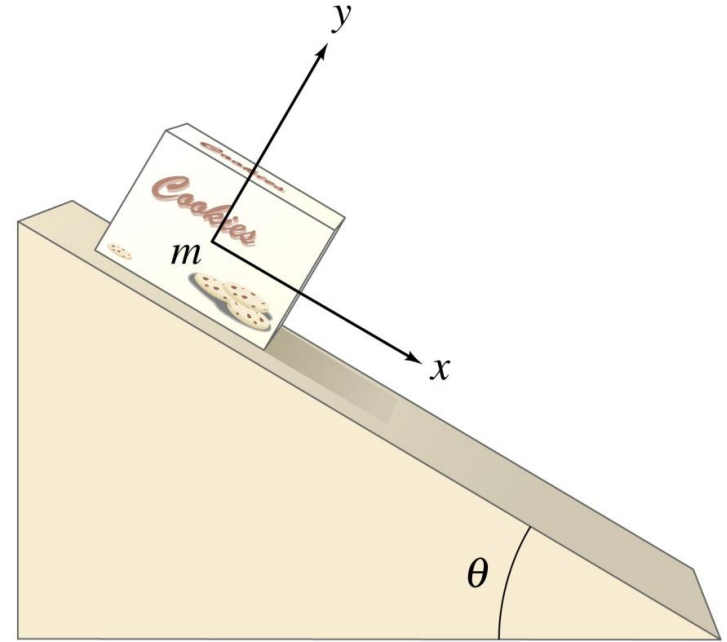


4-6 Friction Force on Inclines

Example 5

The carton lies on a plane tilted at an angle $\theta = 22.0^\circ$ to the horizontal, with $\mu_k = 0.12$.

(a) Determine the acceleration of the carton as it slides down the plane.



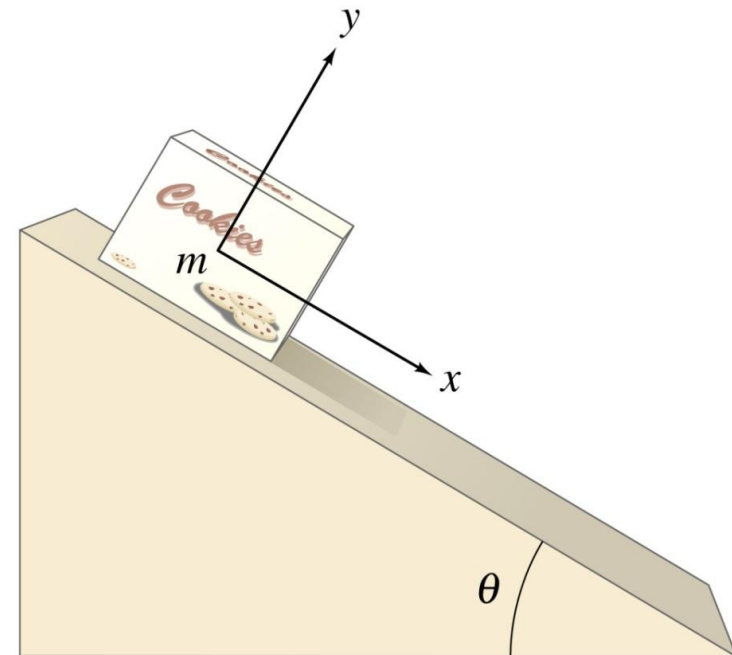
$$\begin{aligned}\sum F_x &= F_{Gx} - F_{fr,k} = ma \\ ma &= mg \sin \theta - \mu_k (mg \cos \theta) \\ a &= g(\sin \theta - \mu_k \cos \theta) \\ &= 9.8(\sin 22.0^\circ - 0.12 \cos 22.0^\circ) \\ &= 2.58 \text{ m.s}^{-2}\end{aligned}$$

4-6 Friction Force on Inclines

Example 5

(b) If the carton starts from rest 9.30 m up the plane from its base, what will be the carton's speed when it reaches the bottom of the incline?

$$\begin{aligned}v^2 &= v_0^2 + 2a\Delta x \\&= 0 + 2(2.58 \text{ m.s}^{-2})(9.30 \text{ m}) \\v &= 6.9 \text{ m.s}^{-1}\end{aligned}$$



4-6 Friction Force on Inclines

Example 5

(c) The carton is now given an initial speed of 3.0 m/s up the 22.0° ramp. If friction is ignored, how far up the plane will it go?

First need to find the net acceleration:

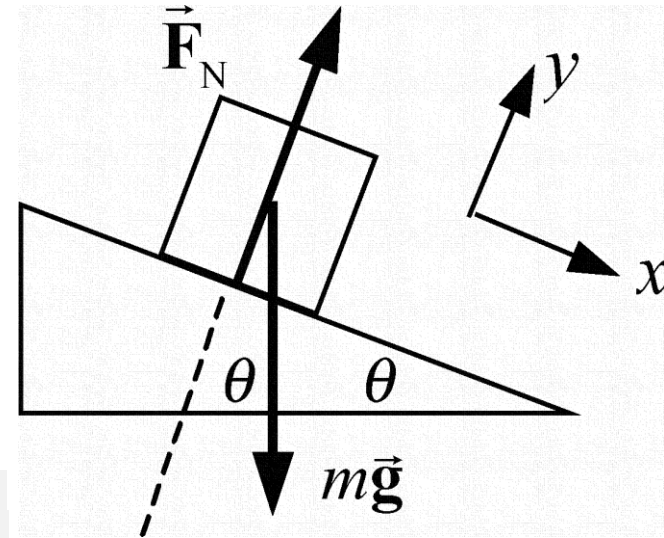
$$\sum F_x = ma \quad \Rightarrow \quad mg \sin \theta = ma$$

$$a = g \sin \theta = 9.8 \sin 22.0^\circ = 3.67 \text{ m.s}^{-2}$$

$$v^2 = v_0^2 + 2a\Delta x$$

$$0 = (3.0 \text{ m.s}^{-1})^2 + 2(-3.67 \text{ m.s}^{-2})\Delta x$$

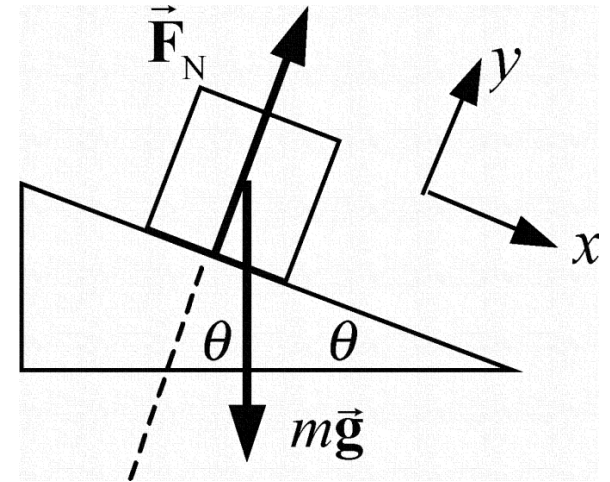
$$\Delta x = 1.2 \text{ m}$$



4-6 Friction Force on Inclines

Example 5

(d) How much time elapses before it returns to its starting point? (Ignore friction)



$$\Delta x = v_0 t + \frac{1}{2} a t^2$$

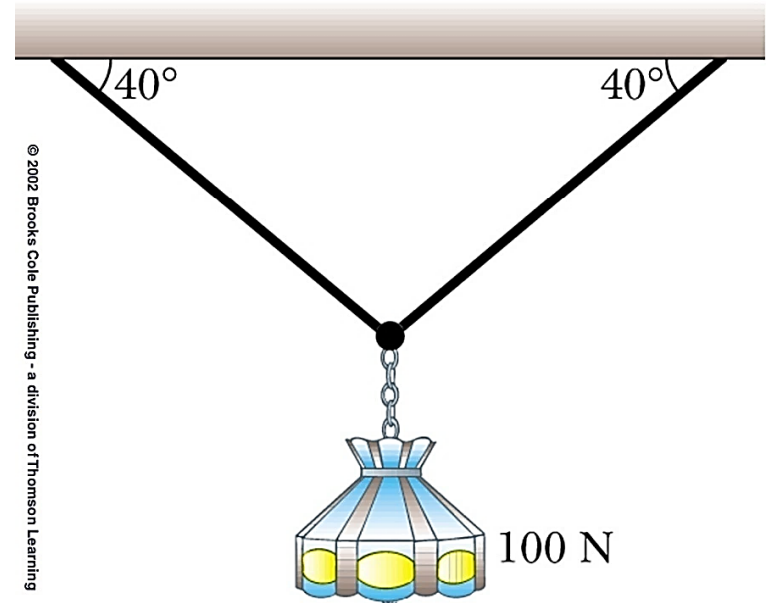
$$0 = (3.0 \text{ m.s}^{-1})t + \frac{1}{2} (-3.67 \text{ m.s}^{-2})t^2$$

$$t = 1.63 \text{ s}$$

4-7 Additional Examples

Example 6

Find the tension in the two wires that support the 100 N light fixture.



In x -direction:

$$-T_1 \cos 40^\circ + T_2 \cos 40^\circ = 0$$

$$T_1 = T_2$$

It is shown that $T_1 = T_2$

In y -direction:

$$T \sin \theta + T \sin \theta - mg = 0$$

$$2T \sin \theta = mg$$

$$T = (100 \text{ N})/2 \sin 40^\circ \\ = 78 \text{ N}$$

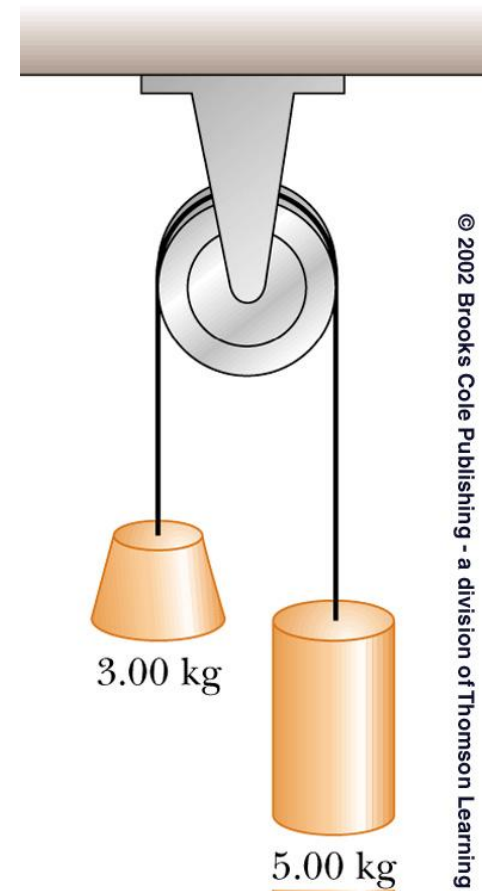
4-7 Additional Examples

Example 7

Two objects with masses of 3.00 kg and 5.00 kg are connected by a light string that passes over a frictionless pulley.

Determine:

- (a) the tension in the string**
- (b) the acceleration of each object**
- (c) the distance each object will move in the first second of motion if both objects start from rest**



4-7 Additional Examples

Example 7

(a) the tension in the string

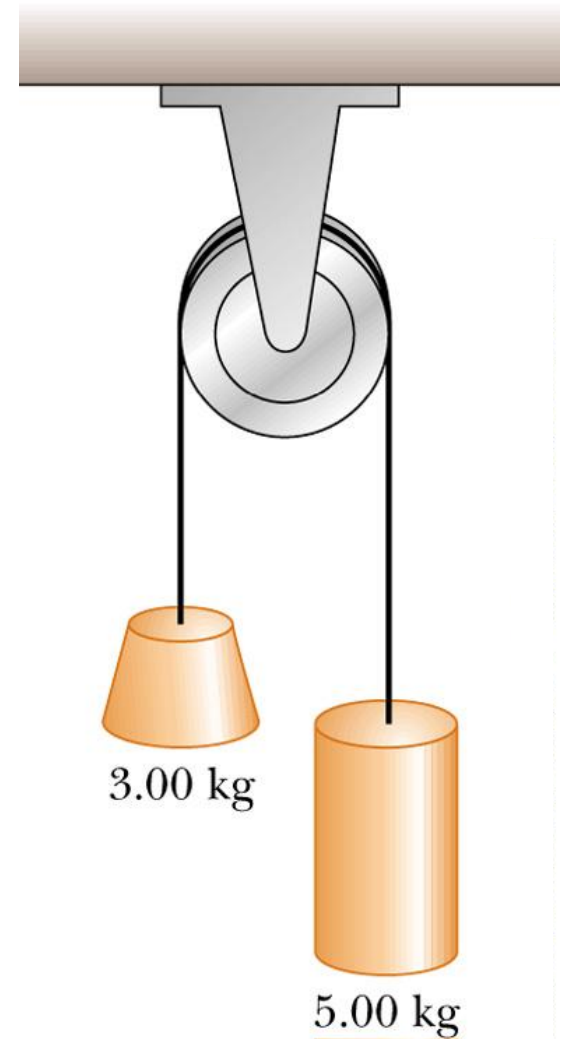
$$T_1 - 3.00g = 3.00a \quad \dots (i)$$

$$5.00g - T_1 = 5.00a \quad \dots (ii)$$

$$\frac{(i)}{(ii)} : \frac{T_1 - 3.00g}{5.00g - T_1} = \frac{3.00a}{5.00a}$$

$$5.00(T_1 - 3.00g) = 3.00(5.00g - T_1)$$

$$T_1 = \frac{30.00g}{8} = 36.8 \text{ N}$$



4-7 Additional Examples

Example 7

(b) the acceleration of each object .

Use Eq. (i):

$$T_1 - 3.00g = 3.00a \quad \dots (i)$$

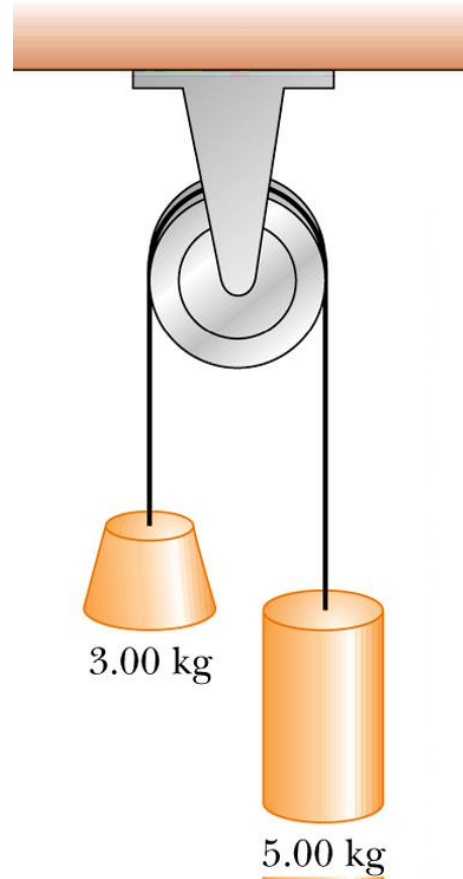
$$36.75 - 3.00(9.80) = 3.00a \quad \Rightarrow a = 2.45 \text{ m/s}^2$$

OR

Consider as one system:

$$\sum F_y = ma \quad \Rightarrow 5.00g - 3.00g = (5.00 + 3.00)a$$

$$2 \times 9.80 = 8.00a \quad \Rightarrow a = 2.45 \text{ m/s}^2$$



4-7 Additional Examples

Example 8

Three objects are connected by light strings. The string connecting the 4.00 kg object and the 5.00 kg object passes over a light frictionless pulley.

Determine:

- (a) the acceleration of each object.
- (b) the tension in the two strings.



Students Practice Questions (III)

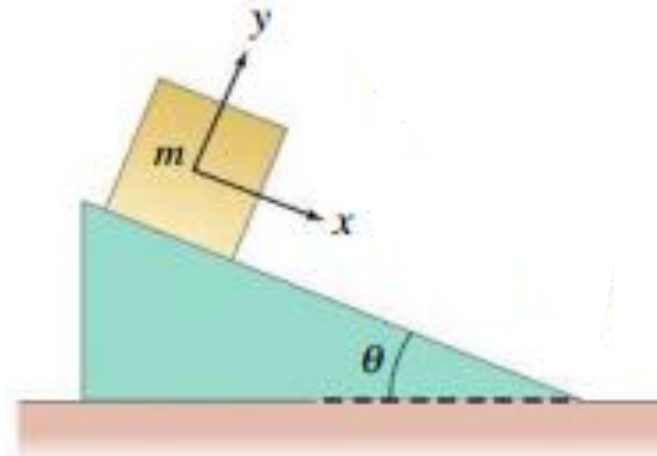
A 2.00 kg block is given an initial speed of 2.50 m/s up a plane inclined 15.0° with the horizontal. The coefficient of kinetic friction between block and plane is 0.250

Draw a free body diagram of the block

- (a) as it is moving up the incline.
- (b) as it is moving down the incline.

Find the acceleration of the block during

- (c) its upward motion on the incline.
- (d) its downward motion on the incline.
- (e) How far does it move up the incline?
- (f) What speed does it have just as it returns to its initial position?



4-6 Problem Solving – A General Approach

- 1. Read the problem carefully; then read it again.**
- 2. Draw a sketch, and then a free-body diagram.**
- 3. Choose a convenient coordinate system.**
- 4. List the known and unknown quantities; find relationships between the knowns and the unknowns.**
- 5. Estimate the answer.**
- 6. Solve the problem without putting in any numbers (algebraically); once you are satisfied, put the numbers in.**
- 7. Keep track of dimensions.**
- 8. Make sure your answer is reasonable.**